

Reviewed Preprint

v1 • December 15, 2025

Not revised

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Reviewing editor: Marcelo U Ferreira, University of São Paulo, Brazil

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Subnational tailoring of ITN distributions to maximise malaria control

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eLife Assessment

This paper describes a **useful** Bayesian model to estimate the probabilities of access, use, and use given access of insecticide-treated bed nets (ITNs), by using sub-national cross-sectional survey data and the annual number of ITNs received at the country level. The authors provide **convincing** evidence to support their modeling approach, which could be enhanced by more validation and exploration of model assumptions.

<https://doi.org/10.7554/eLife.108745.1.sa3>

Abstract

Insecticide-treated nets (ITNs) are the most impactful and cost-effective control tool against malaria. ITNs are primarily distributed through triennial mass campaigns across Africa, though overall ITN use remains modest in many areas as most ITNs do not last three years. In times of funding constraints and a lack of economic alternative antimalarial interventions it is unclear whether disease control could be best improved by distributing more effective ITNs (e.g. dual active-ingredient ITNs) and/or deploying nets more frequently. There are increased calls to improve allocation of resources through sub-national tailoring of interventions, though benefits will depend on how long people use ITNs and how this varies between regions. Here we estimate systematic differences in use and access of ITNs sub-nationally for six countries, Burkina Faso, Ghana, Malawi, Mali, Mozambique and Senegal. These estimates are used to calibrate a *Plasmodium falciparum* transmission dynamics model to generate sub-national estimates of ITN use and cases averted under different ITN distribution strategies. On average, people use their ITNs for 21 months, though this varies substantially between regions from 12 to 38 months. Shifting from triennial to biennial campaigns is estimated to lead to mean population use across all regions increasing from 41.7% to 49.6%. No regions of the 146 investigated were estimated to maintain use over 80% even under biennial distribution, though switching to dual active-ingredient ITNs would likely avert more cases under present distribution frequencies. The framework highlights how routinely collected data can aid policymakers in tailoring disease control programmes at sub-national levels.

Introduction

The widespread use of insecticide-treated nets (ITNs) has prevented more malaria than any other intervention. National malaria programmes (NMPs) have historically aimed to achieve “universal coverage” of ITNs in populations at risk of malaria. This was formally defined by the World Health Organization (2013b) [as 80% of people having access to and sleeping under an ITN the previous night](#) (Roll Back Malaria Partnership, 2005 [as 80% of people having access to and sleeping under an ITN the previous night](#) (Roll Back Malaria Partnership, 2008 [as 80% of people having access to and sleeping under an ITN the previous night](#)). The World Health Organization (WHO) recommended achieving universal coverage by implementing mass distribution campaigns, and through the use of complimentary continuous distribution channels;

these are typically conducted via antenatal care (ANC), the Expanded Programme for Immunization (EPI) and through school-based distribution (World Health Organization, 2024b [↗](#); Koenker et al., 2022 [↗](#)). Mass campaigns have historically been recommended to be conducted every three years with a defined number of nets distributed per person (World Health Organization, 2024b [↗](#); Koenker et al., 2022 [↗](#)). These policies have increased estimated ITN use across sub-Saharan Africa from 2% to 49% between 2000 to 2022 (Bertozzi-Villa et al., 2021 [↗](#)). Access to a net (assuming an ITN could be used by up to two people) has also gone up and 3% to 56% over the same period. While this falls short of the original recommendation, ITNs have been instrumental in reducing the global malaria burden, having averted an estimated 450 million cases between 2000 and 2015 (Bhatt et al., 2015 [↗](#)), or two-thirds of all cases averted through preventative interventions over the same period. Nevertheless, the disparity between recent estimates of ITN use and the original 80% targets are stark. High population use may be unachievable with the status quo of triennial campaigns, since it is thought people do not use their nets for three years, with national median ITN retention times being lower in 87.5% of sub-Saharan African countries (Bertozzi-Villa et al., 2021 [↗](#)). There is also likely to be substantial subnational variability due to environmental and behavioural differences, though evidence is lacking due to the sporadic nature of net surveillance activities. Since 2019, WHO good practice statements have shifted from advocating for ‘universal’ to ‘optimal’ coverage guided by “an explicit prioritization process guiding resource allocation decisions” (World Health Organization, 2024b [↗](#)). However, it is unclear what the optimum coverage is, how it can be estimated, and how this might vary with net retention time.

The epidemiological impact of mass distributions of ITNs will depend on how long individuals retain and sleep under nets and how effective ITNs are at killing mosquitoes and inhibiting blood-feeding. The original rationale for triennial mass campaigns is unclear and is likely based on the durability of first-generation ITNs treated solely with pyrethroid-class insecticides (World Health Organization, 2005 [↗](#), World Health Organization, 2013a [↗](#)). The real-world effectiveness of pyrethroid-based ITNs is under threat from increasingly high levels of pyrethroid resistance in anopheline mosquitoes across sub-Saharan Africa, which is highly heterogeneous over national and sub-national scales (Coleman et al., 2017 [↗](#); Moyes et al., 2020 [↗](#); Hancock et al., 2020 [↗](#), 2022 [↗](#)). Experimental hut trials have indicated increased pyrethroid resistance in the vector population not only reduces the initial mortality conferred by new ITNs, but also increases the rate they lose their insecticidal properties following distribution (Nash et al., 2021 [↗](#); Sherrard-Smith et al., 2022 [↗](#); Churcher et al., 2024 [↗](#)). WHO recommendations suggest that more regular mass campaigns or altering the distribution ratio might be appropriate given local data-driven evidence (World Health Organization, 2024b [↗](#); Koenker et al., 2022 [↗](#)). Nevertheless, international procurers of ITNs, such as the Global Fund, have not historically advised altering the regularity of ITN mass campaigns, nor is this considered in sub-national tailoring exercises, despite this potentially being more cost-effective than layering multiple interventions in the same region.

The World Health Organization has recently expanded its guidance on different types of ITN, with three classes now recommended in areas of pyrethroid resistance. WHO recommendations cascade in the following order: i) pyrethroid-chlorfenapyr nets are strongly recommended for use in place of the standard pyrethroid-only nets; ii) pyrethroid-piperonyl butoxide (PBO) treated nets are conditionally recommended in favour of the standard pyrethroid-only nets, and iii) pyrethroid-pyriproxifen (PPF) treated nets are conditionally recommended in favour of pyrethroid-only nets, but are recommended not to be used in place of pyrethroid-PBO nets (World Health Organization, 2024b [↗](#)). These recommendations are based on results from cluster randomised controlled trials showing that pyrethroid-PBO (Protopopoff et al., 2018 [↗](#); Staedke et al., 2020 [↗](#)) and pyrethroid-chlorfenapyr (Accrombessi et al., 2023 [↗](#); Churcher et al., 2024 [↗](#)) products can both mitigate pyrethroid-resistance associated decreases in ITN effectiveness, while superiority trials for pyrethroid-PPF nets showed mixed results (Accrombessi et al., 2023 [↗](#)). The durability of different classes of nets is also thought to vary substantially, further exacerbating the need to consider the optimum regularity of mass campaigns.

NMPs therefore face two key challenges for planning future ITN interventions. Firstly, to increase ITN use, more frequent mass campaigns may be necessary, given ITNs are typically retained for shorter durations than the status quo mass campaign interval of three years. Secondly, in Africa, where pyrethroid resistance is widespread, newer, yet costlier ITN products are advisable. Addressing either of these challenges would require increases in funding, yet budgets are increasingly under threat and there are increasing calls to tailor ITN distribution within countries to prevent waste and maximise benefit. This is recommended by the WHO (World Health Organization, 2024a [↗](#)) but has not yet been implemented at scale. The impact of both these policy choices may not only depend on how access to ITNs, pyrethroid resistance and transmission intensity vary between and within countries, but may also depend on how likely individuals with access to ITNs are to use them, in addition to the duration that ITNs continue to be used. It is unclear which of these factors are likely to be most associated with improvements in health outcomes through more regular mass campaigns, or more effective ITNs. Moreover, it is unknown if net retention varies systematically within a country or if ITNs remain in use for shorter durations than they are retained for.

To investigate optimal ITN deployment, we estimate the mean duration of ITN access and use from Demographic and Health Survey (DHS) data for different subnational regions in six African countries (Burkina Faso, Ghana, Malawi, Mali, Mozambique and Senegal). Here, regions are defined as the first administrative unit below the country level and are further divided into rural and urban areas. A bespoke mechanistic statistical model is fitted to subnational DHS data from 2005 to 2024. It allows regions with sporadic DHS surveys to be directly compared and generates estimates of ITN use, access and use given access over time, in addition to the contribution towards these by ITNs distributed through continuous channels (ITNs obtained through the private sector, any school- or community-based distribution programme, or through routine ANC and EPI channels). Projections are made for how these metrics change from shifting from triennial to biennial mass campaigns. Using these subnational estimates we calibrated individual-based models of *Plasmodium falciparum* malaria transmission to generate subnational projections of clinical cases averted over 2025 to 2030 under different ITN distribution strategies. Our findings should enable assessment of the epidemiological benefit of different ITN strategies and provide data-driven evidence to improve ITN prioritisation guidance.

Results

Retention time and duration of use

The model estimates people keep and use ITNs for considerably less than three years. Across all regions in all six countries studied, the population-weighted mean duration of ITN access, herein referred to as ITN retention time, was 26.8 months (95% CrI: 26.7, 26.9). Retention times were estimated to vary notably between and within countries ([figure 1 ↗](#)), with median estimates ranging from 14.8 months (95% CrI: 14.5, 15.1) in rural Tete, Mozambique, to 41.5 months (95% CrI: 40.7, 42.2) in rural Ségou, Mali, with only 23 of 146 (15.8%) regions estimated to have a mean retention time greater than three years. The mean durations of ITN use similarly varied notably both subnationally and between countries and were estimated to be consistently lower than mean retention times ([figure 1 ↗](#)), with a population-weighted mean across all regions of 21.0 months (95% CrI: 20.9, 21.0). The mean duration that ITNs remain in use was also estimated to be shortest in rural Tete with a median estimate of 12.0 months (95% CrI: 11.8, 12.3). Meanwhile, the longest mean duration of use was in urban Est, Burkina Faso ([figure 5 – figure supplement 1.A ↗](#)) at 37.8 months (95% CrI: 34.9, 40.6), which was the only region (0.7%) with a median estimate greater than three years. Caution is needed when comparing retention times between countries, due to differences in the timings of surveys relative to mass campaigns, but these are likely to be minimised over subnational scales within a country. While no systematic differences in the durations of use or access were estimated between urban and rural regions in Senegal, differences were observed in some other countries, such as in Ghana, where values were estimated to be

broadly lower in urban settings (figure 5 – figure supplement 2.A [↗](#)). For most regions, the gap between use and access was estimated to widen over time following mass campaigns, indicating people stop using ITNs substantially before they

Universal coverage: was it achievable under triennial mass campaigns?

Access and use of ITNs are estimated to have varied substantially over time, both within and between countries (figure 2.A [↗](#) and 2 - figure supplement 1.A [↗](#)). Inferring changes over time from observational data alone can give misleading results. For example, in rural Centre-Nord in Burkina Faso, DHS survey data suggests a sustained decrease in use between 2014 and 2022 (figure 2.A [↗](#)); however, the model which considered the timings of mass campaigns and routine ITN distribution estimates that overall ITN use immediately following mass campaigns increased over this period. This highlights the potential limitations of making direct comparisons between different surveys, which may be conducted at different intervals since the preceding campaign (figure 2.A [↗](#)). This waxing and waning dynamics in use over time is also observed in *P. falciparum* prevalence (figure 2.B [↗](#)). In many countries peak ITN use and access following mass campaigns has increased over time; for example in rural Thiès in Senegal, the overall use immediately following the first 2009 mass campaign was estimated to reach 18%, yet peak levels of use above 80% were estimated to be achieved following all mass campaigns since 2016 (figure 2 - figure supplement 1.A [↗](#)). Similar trends can be seen in the context of access (dotted lines, figure 2.A [↗](#) and figure 2 - figure supplement 1.A [↗](#)).

Mean ITN use over the three years following a campaign is estimated to be modest, with a population-weighted mean of 41.7% (95% CrI: 39.6, 43.7) across all countries studied. Overall access to an ITN was estimated to be greater over the three year period, with 58.0% (95% CrI: 57.9, 58.1) of people having access to an ITN, on average. The population-weighted mean use given access across all regions was estimated to be 71.9% (95% CrI: 68.2, 75.3). While no regions were estimated to have a mean use or access greater than 80% over either the three-year or two-year periods following a mass campaign, in the context of use given access, 46% of subnational regions had median estimates greater than 80% over a three-year period following a mass campaign, increasing to 70% of regions when only the first two years were considered (figure 3 [↗](#) and figure 3 – figure supplements 1–3 [↗](#)). Nevertheless, some regional differences are observable, with southern Malawi having a low overall ITN use despite the region having a similar level of access to the rest of the country. Across all regions, mass campaigns are estimated to increase ITN use in their immediate aftermath to greater than 80% in 76.8% of regions (bottom row, figure 3 [↗](#) and figure 3 – figure supplements 1–3 [↗](#)), with considerable variability within and between countries. For example, for rural areas in Burkina Faso, the northern Sahel region is predicted to never achieve ITN use above 80%, whereas use is predicted to be maintained above 80% for 6 months following a mass campaign in the south-western region of Cascades.

The contribution of ITNs from continuous channels also differed between and within countries; for example, in rural regions of Thiès (figure 2 - figure supplement 1 [↗](#)) the use of ITNs from continuous channels was estimated to be 15.8% (95% CrI: 14.7, 17.0) at the end of 2024, in comparison to 4.1% (95% CrI: 3.8, 4.5) in Sédhiou. Differences in the use of continuously-distributed ITNs were also notable between countries, with 19.8% (95% CI: 17.4, 22.2) of person-nights of ITN use (one person using an ITN for one night) across Burkina Faso estimated to be attributable to continuously-distributed ITNs over a three-year period, compared to Malawi at 61.2% (95% CI: 58.3, 64.1) (figure 2 - figure supplement 2.B [↗](#)).

There is considerable variability in observed mean access and use within a region, with notable uncertainty between communities, as indicated by the high uncertainty around the probability an individual randomly selected from the region will use an ITN (figure 1.A [↗](#)). The effect of this is illustrated by rural Thiès and Ziguinchor, which both have a mean overall ITN use of 58% when averaged over three years following the last mass campaign. Though the average is the same across both regions, more people in rural areas of Ziguinchor either have a very low or very high

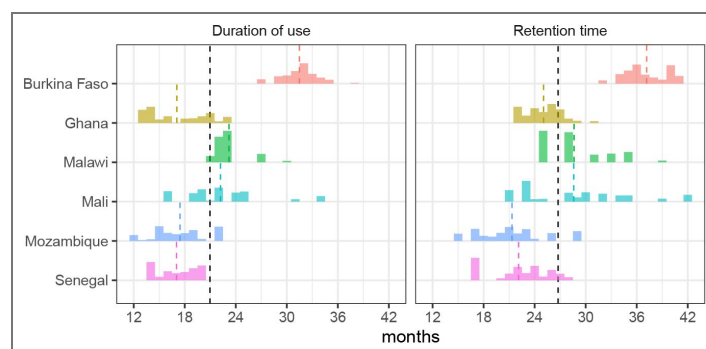


Figure 1. Population-weighted distribution of expected ITN retention times and duration of use by country.

Median estimates of the mean ITN retention time and duration of use for each subnational region were weighted by population to produce country-level distributions. These histograms represent probability distributions of retention time and duration of use for an individual randomly selected from each country, assuming model-derived median estimates for all subnational regions. Subnational variability in retention times and the duration of use reflects differences in model-derived estimates between regions, since the model assumes these values are constant within regions. Vertical dashed lines indicate the overall population-weighted mean (black) and the population-weighted mean for each country (coloured). stop owning them, so that use given access is not constant but typically wanes as ITNs age (figure 2.A [figure 2.A](#)).

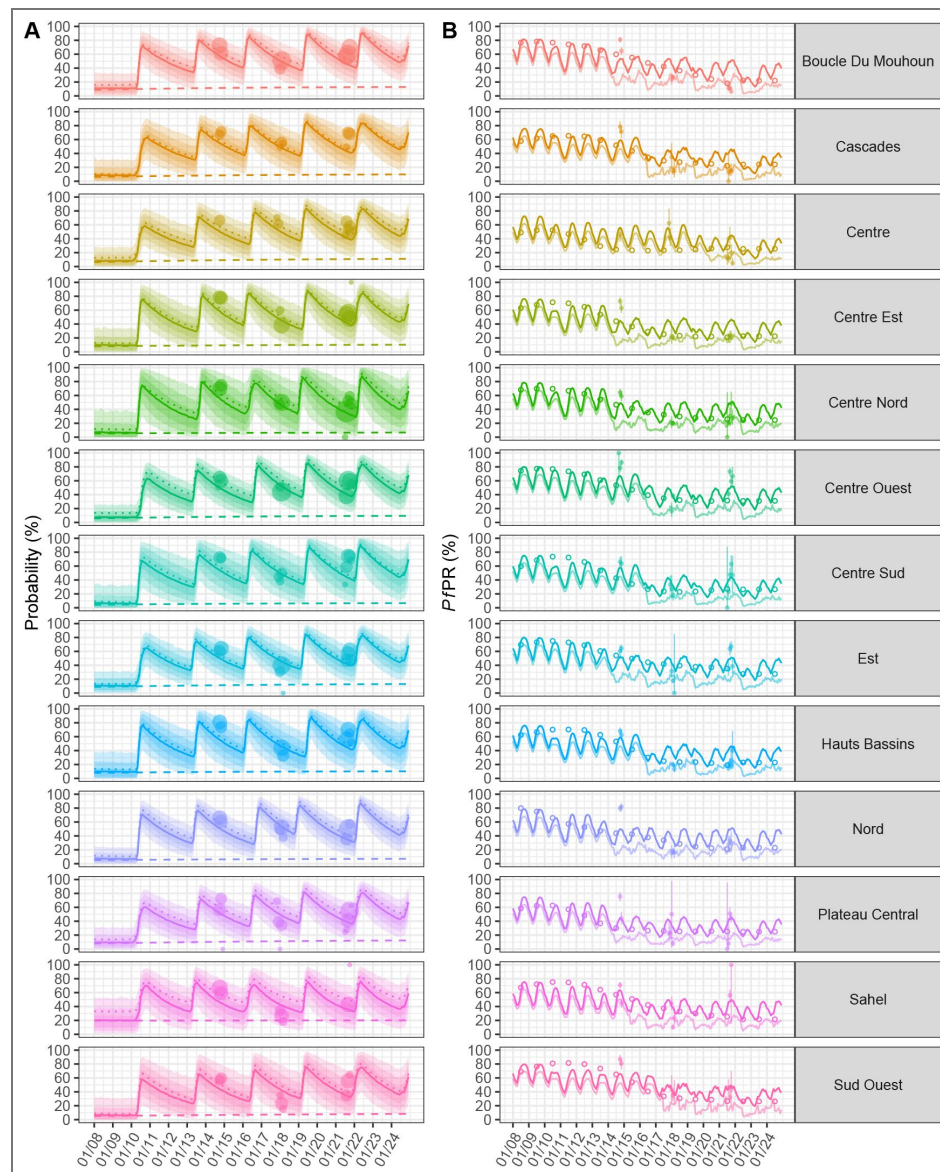


Figure 2. ITN use, access and *Pf* PR over time in rural Burkina Faso.

A: Survey estimates of ITN use (probability someone slept under an ITN the previous night, points sized proportionally to sample size) and model mean estimates of the probability of use over time (solid lines) for rural portions of subnational regions in Burkina Faso. Mean estimates are shown for the probability of use over time (solid lines) and the probability of using an ITN sourced from a continuous channel (dashed lines). The model accounts for differing probabilities of use between individuals within a region; this results in notable uncertainty around the probability an individual selected at random uses an ITN. This uncertainty is illustrated in the shaded region, with the 50%, 80% and 95% credible intervals as indicated by progressively lighter shaded regions. **B:** *Pf* PR₂₋₁₀ estimates from the transmission dynamics model (darker lines) were fitted to annual estimates of *Pf* PR₂₋₁₀ from the Malaria Atlas Project (2024) (hollow points); model estimates of *Pf* PR_{6-59mo} are also shown (lighter lines), in addition to observed DHS *Pf* PR_{6-59mo} and associated 95% credible intervals due to measurement uncertainty (solid points and vertical lines). Figure 2—figure supplement 1. Use, access and *Pf* PR over time in rural Senegal. Figure 2—figure supplement 2. Use of ITNs from continuous channels.

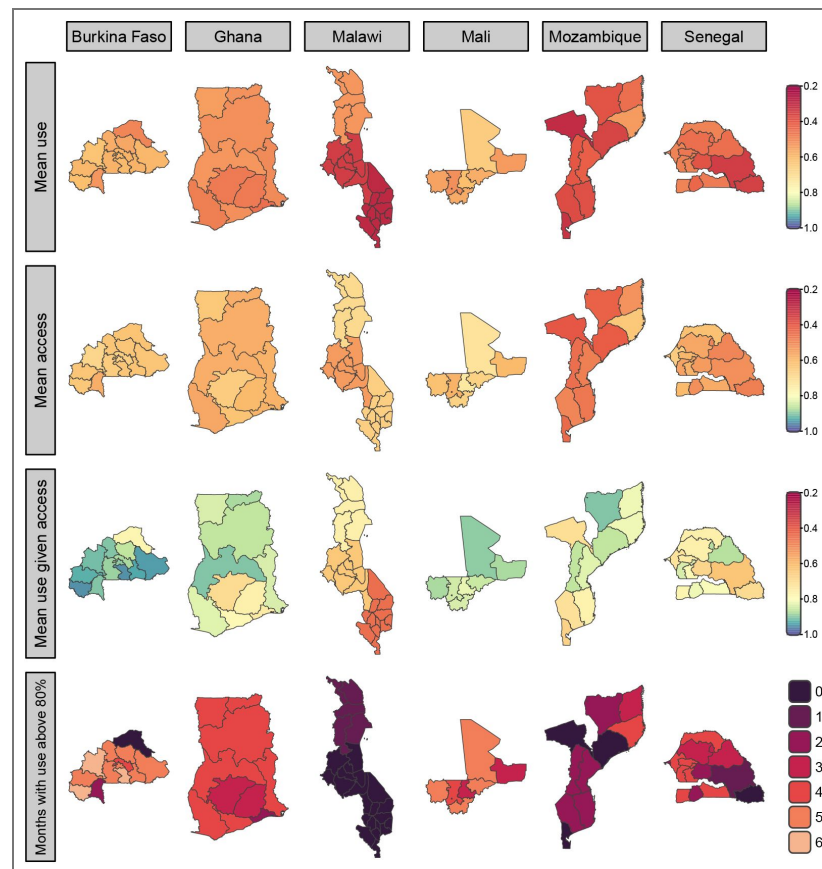


Figure 3. Mean use and access with three-year campaigns in subnational rural areas.

Central estimates of mean overall proportion of people using an ITN the previous night, access to an ITN, and use given access for three-year mass campaign intervals are shown in the top three rows. The number of months following a mass campaign where overall ITN use exceeds 80% are shown in the bottom row. [Figure 3—figure supplement 1](#). **Mean use and access with three-year campaigns in subnational urban areas.** [Figure 3—figure supplement 2](#). **Mean rural use and access with two-year campaigns.** [Figure 3—figure supplement 3](#). **Mean urban use and access with two-year campaigns.**

probability of using an ITN, indicating greater inequity in ITN use in rural areas of Ziguinchor than in Thiès (figure 4 [↗](#)). A similar trend is also apparent in the context of access and in other settings (figure 4 - figure supplement 1 [↗](#) and 2 [↗](#)).

Improving ITN use

The framework can identify regions of low ITN use and make suggestions about how this could be improved (figure 5.A,B [↗](#)). ITNs will be most effective in areas where people have access to nets for a long period of time and are more likely to use them (quadrant 1, figure 5.B [↗](#)). In areas where people keep ITNs but have low use given access (quadrant 2, figure 5.B [↗](#)) NMPs may wish to consider social and behaviour change (SBC) interventions such as ‘hang-up’ campaigns. Conversely, in areas with lower ITN retention times but high use given access (quadrant 4, figure 5.B [↗](#)) NMPs may wish to consider more regular mass campaigns to improve equality of access between different regions. Both of these policy changes may be warranted in regions with both low retention and use given access to improve overall ITN use (quadrant 3, figure 5.B [↗](#)). Results varied by country (figure 5 - figure supplements 1-5 [↗](#)).

The statistical model can be used to predict the increase in use caused by increasing the frequency of mass campaigns from three to two years, assuming the contribution of continuous distribution channels and the decay in use given access following mass campaigns remains the same. Increases in access are estimated to be more modest, with relative population-weighted mean increases across all settings of 14.5% (95% CrI: 14.5, 14.6), from 41.7% to 49.6%. Greater increases are predicted to be seen for ITN use, with mean use across all settings increasing from 58.0% to 66.2%, an increase of 19.5% (95% CrI: 19.5, 19.6). Subnational variation is also predicted in changes of access, use and use given access following adoption of biennial mass campaigns (figure 3 [↗](#) and corresponding figure supplements 1-3).

Improving ITN effectiveness

Policy-makers may wish to optimise ITN distribution to maximise epidemiological impact. We generate estimates of clinical cases under different distribution strategies by simulating halting mass campaigns or changing the campaign interval from three to two years. Fifty eight percent of ITNs distributed in sub-Saharan Africa in 2023 were reported to have been pyrethroid-PBO nets (World Health Organization, 2024c [↗](#)); we therefore estimate changes in clinical cases for alternative strategies compared to the current triennial pyrethroid-PBO campaigns with supplementary continuous distribution, with triennial pyrethroid-chlorfenapyr strategies as a comparator also presented (figure 6 - figure supplement 1 [↗](#)).

The number of clinical cases averted by the different strategies varies substantially, both within (figure 5.C [↗](#) and figure 5 - figure supplement 1 [↗](#)-5 [↗](#), panel C) and between countries (figure 6 [↗](#)). For example, in Mali switching from a triennial to biennial pyrethroid-PBO mass campaign is estimated to avert fewer than 50 additional cases per 1,000 people in Gao, but over 100 in Sikasso (figure 5.C.vi [↗](#)). The benefit of the different strategies is projected to differ in each setting, though broad trends are observable. Generally, fewer cases are projected to be averted under triennial pyrethroid-chlorfenapyr campaigns than with biennial pyrethroid-PBO distribution, with 74.6% of regions having fewer clinical cases predicted when you increase the campaign frequency than when the net class is changed (figure 6 - figure supplement 2.F [↗](#) vs. H and regions with negative y-axis values in figure 6 - figure supplement 1.F [↗](#)). Triennial distribution of pyrethroid-chlorfenapyr ITNs was predicted to avert more cases than biennial distribution of pyrethroid-only ITNs for 94.3% of regions (figure 6 - figure supplement 2.H [↗](#) vs. C and regions with positive y-axis values in figure 6 - figure supplement 1.C [↗](#)). This was more consistent for policy decision-making than the differences between triennial pyrethroid-PBO distribution and biennial pyrethroid-only distribution, with distributing pyrethroid-PBO ITNs averting more cases in 65.6% of regions (figure 6 - figure supplement 2.E [↗](#) vs. C and regions with positive y-axis values in figure 6.C [↗](#)).

It is unlikely that bespoke simulation modeling will be available for all policy decisions, but the models can indicate trends that can support sub-national tailoring of ITNs. The models indicate current clinical incidence is the best predictor of the number of cases averted for the different

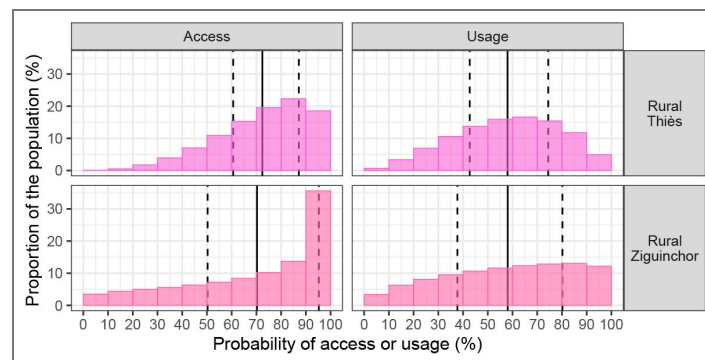


Figure 4. Equity of use and access in rural Thiès and Ziguinchor, Senegal.

Solid vertical lines denote overall ITN use and access averaged over three years following the last mass campaign. Coloured bars indicate the proportion of the population with different probabilities of using or having access to an ITN. For example, 10% of the population have a probability of access between 70% and 80% in rural Ziguinchor, in comparison to 20% in rural Thiès. Vertical dashed lines denote 50% credible intervals for the probability of an individual uses or has access to an ITN, indicating that half of the population are expected to have a probability of use or access within this range. [Figure 4—figure supplement 1](#). **Subnational equity of use in Senegal.** [Figure 4—figure supplement 2](#). **Subnational equity of access in Senegal.**

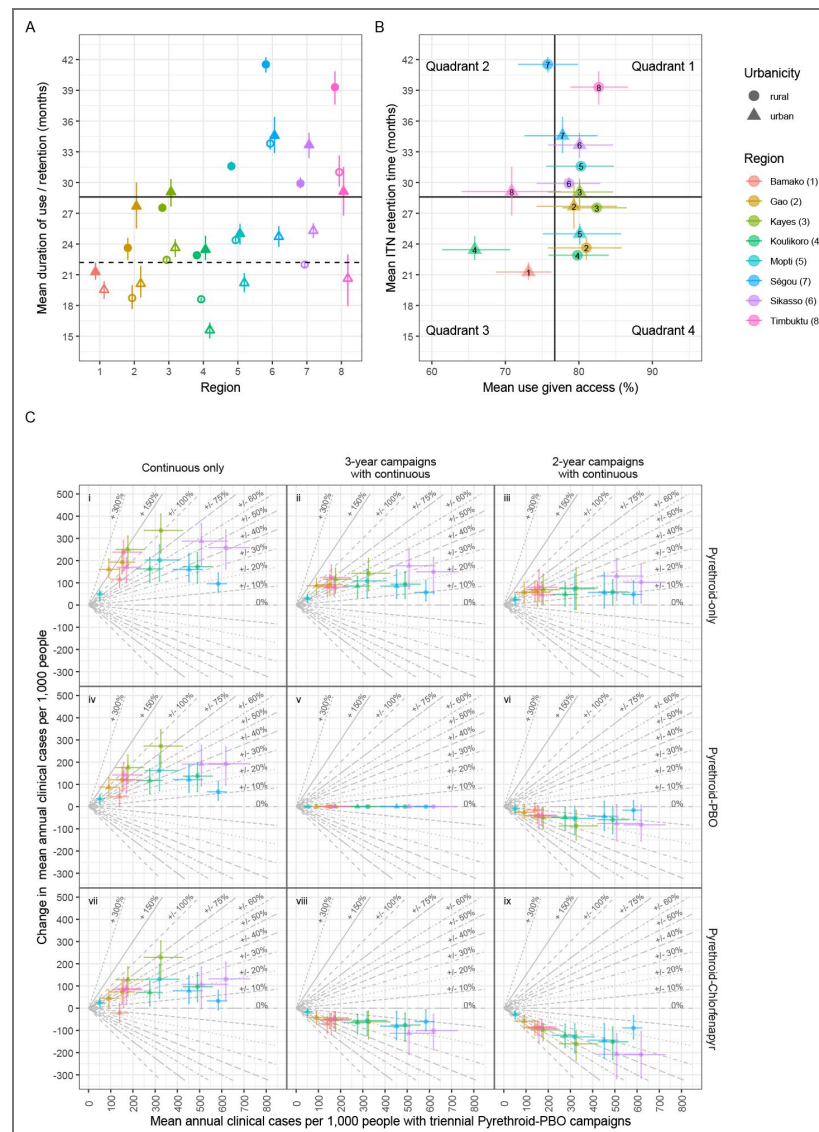


Figure 5. Retention time, use given access and changes in cases for Mali.

All coloured points indicate median estimates and associated 95% credible intervals for subnational regions in Mali. **A:** Mean durations of ITN use (hollow) and retention (filled) are shown in addition to national median estimates (black solid and dashed lines, respectively). **B:** Mean ITN retention and use given access under a triennial campaign strategy are shown, with black lines indicating national median estimates. **C:** Points indicate mean annual clinical cases under a triennial pyrethroid-PBO campaign strategy, and the projected change in cases for alternative intervention strategies with pyrethroid-only (**C.i-iii**) pyrethroid-PBO (**C.iv-vi**) or pyrethroid-chlorfenapyr (**C.vii-ix**) ITNs, with continuous-only distribution (**C.i, iv, vii**), or in conjunction with triennial (**C.ii, v, viii**) or biennial (**C.iii, vi, ix**) campaigns. The change in clinical cases is equal to zero for equivalent comparator and intervention strategies (**C.v**). Labeled diagonal reference lines with positive gradients indicate percentage increases in clinical cases relative to mean annual estimates under the comparator strategy; lines of the same style with negative gradients indicate equivalent percentage decreases. Urban areas were estimated to broadly have lower use given access than in rural settings, while the capital city, Bamako, was estimated to have the lowest ITN retention time of all regions. [Figure 5—figure supplement 1](#). **Retention time, use given access and changes in cases for Burkina Faso.** [Figure 5—figure supplement 2](#). **Retention time, use given access, and changes in cases for Ghana.** [Figure 5—figure supplement 3](#). **Retention time and use given access for Malawi.** [Figure 5—figure supplement 4](#). **Retention time, use given access and changes in cases for Mozambique.** [Figure 5—figure supplement 5](#). **Retention time, use given access and changes in cases for Senegal.**

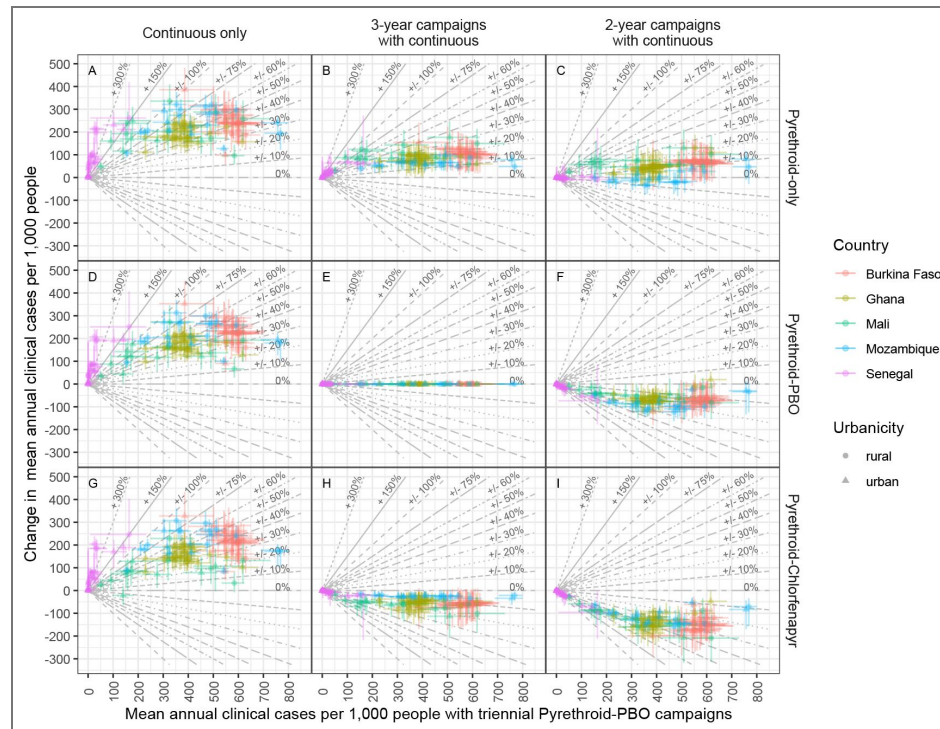


Figure 6. Change in cases vs. cases with triennial pyrethroid-PBO distribution.

Points represent urban and rural areas within each subnational region, are colored by country, and are sized proportionally to the mean annual number of ITNs distributed under each strategy. Each point shows median estimates of the change in clinical cases following a switch from triennial pyrethroid-PBO distribution to alternative strategies against the projected clinical cases under the triennial pyrethroid-PBO strategy. Concurrent distribution of continuous ITNs of the same class used in mass campaigns is assumed throughout. Vertical and horizontal lines indicate 95% credible intervals. Labeled diagonal reference lines with positive gradients indicate percentage increases in clinical cases relative to mean annual estimates under the comparator strategy; corresponding lines with negative gradients represent equivalent percentage decreases and share the same line style. [Figure 6—figure supplement 1](#). **Change in cases vs. cases with triennial pyrethroid-chlorfenapyr distribution.** [Figure 6—figure supplement 2](#). **Cases averted under different ITN distribution strategies.** [Figure 6—figure supplement 3](#). **Change in cases vs *Pf* PR with triennial pyrethroid-PBO distribution.** [Figure 6—figure supplement 4](#). **Change in cases vs pyrethroid resistance.** [Figure 6—figure supplement 5](#). **Change in cases vs duration of use.** [Figure 6—figure supplement 6](#). **Change in cases vs retention time.** [Figure 6—figure supplement 7](#). **Change in cases vs use.** [Figure 6—figure supplement 8](#). **Change in cases vs access.** [Figure 6—figure supplement 9](#). **Change in cases vs use given access.**

strategies. For example, for both pyrethroid-PBO (figure 6.F [↗](#)) and pyrethroid-chlofenapyr ITNs (figure 6 - figure supplement 1.I [↗](#)), the projected additional cases averted by switching to biennial distribution is estimated to gradually increase as clinical cases increase up to approximately 300 mean annual clinical cases per 1,000 people, though above this point trends are less obvious. Similar trends are seen in countries with a broad range of transmission intensities, such as Mali (figure 5.C.vi [↗](#)) and

Mozambique (figure 5 - figure supplement 4.B.vi [↗](#)), though it is less clear in countries where there are no low transmission regions, such as in Burkina Faso (for example, figure 5 - figure supplement 1.C.vi [↗](#)). Similar trends can be observed by using all-age malaria prevalence as an indicator to select regions which will have the greatest benefit of greater investment in ITNs (figure 6 - figure supplement 3 [↗](#)). Selecting areas to target using the level of pyrethroid resistance is less reliable (figure 6 - figure supplement 4 [↗](#)), in part because there is little variability within countries and differences between countries are confounded by transmission intensity in this analysis.

Surprisingly, there was no discernible association between either ITN retention times or the mean duration that nets remain in use on the impact of increasing campaign frequencies (figure 6 - figure supplements 6.F [↗](#), 5.F [↗](#)). This is because ITNs have a bigger epidemiological impact in areas where they are used for longer, though policy-makers may wish to consider equity in disease burden in their tailoring process which is not considered here. There were, however, clearer relationships that indicated more cases may be averted following a switch from pyrethroid-PBO to pyrethroid-chlofenapyr triennial distribution in regions with greater mean access, use and use given access (figure 6 - figure supplements 8.H [↗](#), 7.H [↗](#), 9.H [↗](#)), though these associations are weaker than those seen for clinical cases or prevalence.

The cessation of mass campaigns is projected to lead to notable resurgences in clinical cases in many regions; an additional 150 to 300 clinical cases per 1,000 people are estimated for moderate to high transmission intensity regions (those with more than 150 mean annual clinical cases per 1,000 people under triennial pyrethroid-PBO distribution strategies). It has been suggested that the potential increase in cases caused by halting mass campaigns could be mitigated by concurrently switching to using pyrethroid-chlofenapyr ITNs in continuous distribution channels (figure 6.G [↗](#) vs. D [↗](#)). This modeling exercise indicates such mitigation is unlikely, with only 1 of the 146 regions showing a reduction of cases when switching from combined continuous distribution and triennial mass campaigns with pyrethroid-PBO to continuous distribution alone with pyrethroid-chlorfenapyr. This was for the capital city of Mali, Bamako (figure 5.C.vii [↗](#)), which has moderately low transmission, while Mali was estimated to distribute more ITNs through continuous channels than most other countries considered in this analysis (figure 2 - figure supplement 2 [↗](#)).

Discussion

International guidelines on the use of ITNs have changed over the past two decades, moving from a universal target of 80% overall use in regions at risk of malaria to more nuanced guidance of “optimal” coverage in areas with resource limitations (World Health Organization, 2024b [↗](#)). This was an important policy change as it allowed NMPs to consider the use of alternative interventions which may be more cost-effective than pushing for very high population use in areas where this might be unachievable. Our work supports this policy change as it shows for the first time how ITN use consistently varies subnationally between regions in all countries where DHS data were available to test the hypotheses. Nevertheless, high population use of ITNs is immensely effective at preventing malaria, with higher population use leading to better outcomes. The modelling results indicate that the overall 80% threshold was seldom consistently achieved across many countries in sub-Saharan Africa, and though this threshold appears relatively arbitrary, it is unclear how the new guideline of ‘optimal’ ITN use should be assessed. Mean use of ITNs under triennial mass campaigns with supplementary continuous distribution was estimated to be, on average, 41.7% (95% CrI: 39.6, 43.7) over all regions included in our analysis. This highlights that status quo approaches are typically yielding mean levels of ITN use that are

roughly half of minimum universal coverage targets. Unless more nets are distributed or people keep their nets for significantly longer, universal coverage has potentially always been an unobtainable goal between mass campaigns.

The benefit of ITNs depends on how long people keep and use them. We therefore aimed to identify systematic differences in subnational use of and access to ITNs. Our central estimates indicated that 86.3% of subnational regions had a mean duration of access less than three years. Although the subnational regions included in this study are from a subset of countries in sub-Saharan Africa, our findings are broadly consistent with previous model-derived estimates of national median retention times, which have been estimated to be less than three years for 87% (Bertozzi-Villa et al., 2021 [DOI](#)) to 100% (Bhatt et al., 2015 [DOI](#)) of sub-Saharan African countries. Since we assume the duration of ITN use and access is exponentially distributed, our models incorporate a greater probability that a net will be lost, or cease to be used, in the immediate period after receipt. Therefore, we would argue our mean estimates are more comparable to the median estimates by Bertozzi-Villa et al. (2021 [DOI](#)) and Bhatt et al. (2015 [DOI](#)), which both assume a sigmoidal decay function. Although some field studies have reported ITN retention times greater than three years (Obi et al., 2020 [DOI](#); Diouf et al., 2022 [DOI](#)), results differ by location, with the majority of field studies (for example, Gnanguenon et al., 2014 [DOI](#); Solomon et al., 2018 [DOI](#); Lorenz et al., 2020 [DOI](#)) estimating that ITN retention fails to exceed three years in most settings. Even in isolation, our projections of additional cases averted under biennial over triennial campaigns raise further questions on whether the status quo of three-year intervals is sufficient to maintain adequate or “optimal” coverage; especially given its original basis stems from the duration of pyrethroid-class insecticidal activity estimated two decades ago (World Health Organization, 2005 [DOI](#), World Health Organization, 2013a [DOI](#), 2023 [DOI](#)).

While previous work on net retention has largely focused on access, here we have also estimated the mean duration for which ITNs remain in use, which has greater epidemiological importance. Median estimates of the mean duration of use were less than three years in all but one of the subnational regions investigated. The work highlights that there is a faster rate of loss of use than access. Although the causes of this are unclear, it could be driven by poor quality ITNs, with people with older nets potentially not using them if degraded ITNs are perceived to provide lower protection. Indeed, reduced ITN integrity has been associated with reduced probability of use (Hiruy et al., 2021 [DOI](#)). ITN age is thought to be a better predictor of infection risk than durability alone (Andronescu et al., 2019 [DOI](#)). Therefore, even if duration of use is lower in some regions due to reduced ITN integrity, SBC campaigns either to promote use or to help with net care may still be beneficial.

We characterised historical trends in overall ITN use and access over time in different subnational regions. These estimated a gradual increase in both overall use and access over time, with waxing and waning dynamics driven by timings of mass campaigns; similar national patterns of declining use following mass campaigns have also been estimated in models by (Bhatt et al., 2015 [DOI](#)) and recorded in longitudinal field surveys (Hamre et al., 2020 [DOI](#)) and cluster randomised control trials (Protopopoff et al., 2018 [DOI](#)). Furthermore, surveillance data in multiple settings have also highlighted notable increases in malaria cases in the second year following mass campaigns (Zhou et al., 2016 [DOI](#); Wotodjo et al., 2017 [DOI](#); Girond et al., 2018 [DOI](#); Topazian et al., 2021 [DOI](#)), providing further, albeit indirect, indications that overall ITN use declines notably in real-world settings by the second year following mass campaigns.

Although neither overall ITN use nor access in any subnational regions were projected to be maintained above the universal coverage threshold of 80%, mean use given access under present triennial strategies was estimated to be greater than 80% for almost half of subnational regions. This highlights that two decades after their widespread introduction, ITNs remain a valued commodity and key weapon in the expanding arsenal of malaria interventions when individuals have access to them. Moreover, if DHS surveys were primarily conducted in dry seasons in some locations (which is often the case), when ITN use given access is typically lower, our estimates in turn may be underestimated in some settings, especially in Senegal and regions of Mali, which are

thought to have more seasonal fluctuations in use (Koenker et al., 2019b). However, to achieve consistent attainment of high population use targets, alternative distribution strategies, such as enhanced continuous distribution channels may be needed (Koenker et al., 2023b).

By identifying subnational regions with systematic low use given access and short durations of access, we highlight geographical areas where ITNs are likely to perform less effectively, irrespective of the local transmission intensity. This may aid NMPs in tailoring subnational strategies aside from those related to ITN distribution. For example, in regions where local knowledge indicates low use given access is driven by social and/or behavioural factors, SBC communication may be beneficial. Local understanding of the barriers to ITN use will be instrumental in assessing the costeffectiveness of SBC strategies. While SBC communications, such as ‘hang-up’ campaigns have led to reported 250% relative increases in use in Zambia (Macintyre et al., 2012) and absolute increases between 11% and 17% in Togo (Desrochers et al., 2014) and Nigeria (Kilian et al., 2016), in other settings, such as in Uganda (Kilian et al., 2015), significant increases in use was not observed. We have not investigated the underlying factors that may have led to poor utilisation of ITNs in some subnational regions. We assume that use, access, and use given access immediately following a campaign, as well as their decay rates, remain unchanged when shifting from triennial to biennial strategies, and further work is needed to understand how use may change for different distribution strategies. Importantly, poor utilisation may arise due to ITN degradation (Mboma et al., 2021). An assessment of ITN durability studies across seven sub-Saharan African countries by Briet et al. (2020) found greater variability in durability between locations than between different net brands. Damaged ITNs may contribute not only to reduced utilisation, but also in reduced efficacy, particularly in areas with pyrethroid-resistant mosquitoes (Wheldrake et al., 2024).

Subnational tailoring of ITN interventions is increasingly recommended, with some World Health Organization (2024b,a) guidelines currently advising lower transmission settings to be considered first for deprioritisation from mass campaigns. Optimal subnational tailoring may not solely rely on deprioritisation, but may also require reprioritisation of resources to other regions. Increasing the frequency of mass campaigns or procuring more effective, but likely more expensive, ITNs will inevitably increase the costs of conducting mass campaigns if the same level of coverage is achieved. A thorough economic analysis is needed to understand the best ITN strategy, especially as ITNs remain one of the most cost-effective malaria control interventions so improving the effectiveness of campaigns could be more impactful than the introduction of new untested interventions (Topazian et al., 2023; Schmit et al., 2024). In the context of finite procurement budgets, our framework identified subnational regions where increasing the frequency of mass campaigns or switching to more effective ITNs may have the greatest impact. Although we only included six countries in our analysis, they not only cover a wide geographical range, but also a broad spectrum of *P. falciparum* transmission intensities and estimated levels of pyrethroid resistance (Hancock et al., 2020). For all countries considered, transmission intensity was positively correlated with increased benefit of both switching to two-year mass campaign cycles or more efficacious ITN classes. The framework can be extended to other countries as DHS data becomes more available. In the absence of this, our results indicate areas with greater transmission intensity are likely to see the greatest impact from either increasing mass campaign frequencies or switching to pyrethroid-PBO or pyrethroidchlorfenapyr ITNs, supporting WHO guidelines World Health Organization (2024a). Ultimately, decisions need to be made by NMPs whether they want to target the most cases, or whether decisions should be made with respect to equity of resources (does everyone have similar access to a net) or equity of impact (does everyone have a similar chance of getting malaria), both within and between regions.

As the toolbox for malaria control continues to expand, there is increasing need for ongoing monitoring and evaluation to facilitate evidence-based decision making and optimal deployment of interventions (Wilson et al., 2020). Longitudinal standardised surveys, such as the DHS, have the potential to be instrumental in assessing the effectiveness of interventions in real world settings, and have the potential to be used to refine distribution strategies to maximise impact. Such punctuated surveys can aid in monitoring the use of different public health products, such as

spatial repellents or traps, and could independently assess the success of drug and vaccine campaigns. However, the timing of DHS surveys every few years creates challenges for assessing the effectiveness of both ITNs and other commodities. For example, if two DHS surveys were conducted at different times in relation to the last mass campaign, underlying trends in ITN use or access may fail to be captured. Similarly, if data are collected irregularly in relation to other future punctuated interventions, such as intermittent rapid diagnostic testing following mass vaccination campaigns, the methodology presented here provides a framework where potential biases can be controlled for, provided estimates of the duration of vaccine-induced immunity and the timings of vaccination campaigns are known. With appropriate adaptation to our model, this framework can facilitate more reliable inferences on the effectiveness of interventions beyond malaria than would be possible from survey data alone.

In conclusion, mass campaigns form a key pillar in the effort to ensure effective ITN coverage across countries at risk of malaria. The work indicates that universal coverage targets of 80% are unlikely to be consistently met due to waning overall ITN use in the intervening years between triennial mass campaigns. Improved coverage can be achieved through more frequent biennial distributions, though this is unlikely to be feasible at scale given current funding envelopes. Frameworks such as that presented here - which take into account the potential for impact from different net types and the high variability of ITN duration and use - could support NMP decision-making on how best to maximise impact from available funds. Whilst such frameworks may be a useful tool, local knowledge of factors impacting ITN access and use as well as operational decision making will be paramount for NMP-led tailoring of subnational strategies.

Methods and Materials

Inclusion criteria and data sources

The majority of data were sourced from Demographic and Health Surveys (DHS) and Malaria Indicator Surveys (MIS) (ICF, 2025 [link](#)). AIDS Indicator Surveys (AIDS) were also included where additional malaria questions were included (ICF, 2025 [link](#)). Data on the number of ITNs distributed nationally were also sourced from (The Alliance for Malaria Prevention, 2024 [link](#)).

Here we assume each ITN is used every night for a continuous period after receipt (duration of use), but may remain accessible for a different length of time following receipt (duration of access/retention time). It is important to distinguish between people who have access to an ITN but don't sleep beneath it and those who don't have access to one. We generate subnational estimates of both the population mean duration of use and retention time and assume these are constant over time. Single estimates of both these quantities are estimated separately for rural and urban areas of each region to allow the identification of systematic differences between locations that could allow the geographical targeting of resources.

Retrospective analyses were conducted from 2008, to align with the introduction of mass distribution campaigns, until 2024. Analyses were conducted at an administrative-one level stratified by urban and rural settings, as recorded in survey responses. To account for inconsistencies in administrative-one naming conventions, such as alternative French and English spelling, best efforts were made to unify recorded administrative-one entries with the naming conventions recorded by GADM v4.0 (Global Administrative Areas, 2021 [link](#)). Note that while administrative-one is referred to as a region in Senegal, for example, terminology differs between countries; here, the units of analysis are consistently referred to as subnational regions. The initial inclusion criteria were restricted to countries where at least three DHS/MIS surveys had been conducted between 2010 and 2022. This was to ensure there were sufficient data for all countries after mass campaigns had been established. Ten countries initially met this criteria: Burkina Faso, Ghana, Liberia, Malawi, Mali, Mozambique, Nigeria, Senegal, Tanzania and Uganda.

All mass campaigns were assumed to be conducted nationally, although their timings were not restricted to occur in synchrony across subnational regions. Instead, the timing of each mass campaign was allowed to vary subnationally within time intervals that were defined at a national level. Due to this assumption, Liberia was excluded from our analysis due to rolling campaigns

being conducted between 2008 and 2012 (Koenker et al., 2019a). We also excluded Nigeria where mass campaigns follow different distribution schedules between states (Koenker et al., 2019a). Tanzania was also excluded as a mixture of annual school-based distributions and five-yearly mass campaigns are conducted within different subnational regions (Koenker et al., 2023a). We finally excluded Uganda since there have been notable changes to its administrative-one level boundaries over the assessed period; this has resulted in significant discrepancies between the administrative-one level regions recorded in DHS surveys and the naming conventions used in GADM v4.0. Although Ghana was included in all analyses, the subdivision of the regions of Brong-Ahafo, Northern and Volta in 2019 was not accounted for in the retrospective analyses of historical use and access; the new regions were however accounted for in the transmission dynamics models for future projections. For Malawi, we conducted retrospective analyses of use and access at an administrative-one level (termed regions in the country); however, as data we sourced for other covariates, such as rainfall patterns and non-ITN interventions, were stratified at an administrative-two level (districts) for Malawi (Winskill, 2024), we excluded the country from the transmission dynamics analyses. These other covariates were stratified at the administrative-one level for all other countries (Winskill, 2024).

All subsequent analyses described were conducted in R (v4.3.2, R Core Team, 2023). The code for models used in this study can be found at https://github.com/andrewcglover/itn_campaigns. Unless otherwise stated, median central estimates are quoted in the main text.

Initial estimation of net retention times

We initially sought to estimate subnational retention times of ITNs from DHS and MIS survey data using the age distributions of ITNs in surveyed households. Due to the sparsity and irregularity of DHS and MIS surveys, we were unable to investigate seasonal fluctuations in both access and use; we therefore assume that a net in region i of a total N_i regions is used continuously over some period of time, ρ_i^u , but may provide access for a longer period, ρ_i^a . We also assume that all nets in region i cease to be used and are disposed of by households at constant, but not necessarily equal, rates, γ_i^u and γ_i^a , respectively, such that:

$$\rho_i^x \sim \text{Exponential}(\gamma_i^x) \quad (1)$$

where $x = u$ and $x = a$ in the contexts of use and access, respectively. Under this assumption, the expected mean duration of use and access (the retention time) of nets are therefore $1/\gamma_i^u$ and $1/\gamma_i^a$, respectively.

For the purposes of generating our initial estimates of net retention, we also make additional assumptions on how the crop of ITNs is replenished in the population by distribution strategies. Some of these assumptions are however relaxed in later sections. Here we assume the number of nets present in the population received through continuous distribution channels remains at a constant level n_d by continual replacement of discarded nets. Here, we also initially assume that nets are distributed without replacement and that n_c^0 nets are distributed through mass campaigns every Δt months. Under these assumptions, if N_s surveys are conducted over the interval $[s_0, s_m] \in T$ (figure 7), then the age distribution of ITNs that were used and or provided access over all surveys can be shown to approximately follow the distribution in equation (1). Therefore, given a sufficiently large number of surveys were conducted over a sufficiently long period of time, the ages of used, α_i^u , and accessible, α_i^a , nets in region i approximately follows:

$$\alpha_i^x \sim \lim_{\substack{N_s \rightarrow \infty \\ s_m \rightarrow \infty}} \text{Exponential}(\gamma_i^x) \quad (2)$$

Therefore, provided a mass campaign strategy is suitably long-established where a large number of surveys have been conducted over time, the rate of net loss can be estimated from fitting an exponential distribution to the empirical age distribution of ITNs across all recorded DHS surveys. Additionally, the limit behaviour in equation (2) can be shown to hold both when surveys are conducted randomly or at regular intervals Δs , provided that $\Delta s \neq \Delta t$ (figure 7).

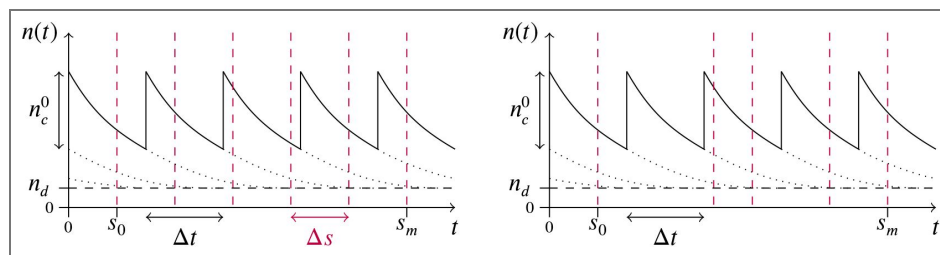


Figure 7. An illustration of the number of nets, $n(t)$, in a region over time with continuous replenishment of routinely distributed nets, n_d , where n_c^0 nets are distributed every Δt years through regular mass campaigns (solid black lines).

The contribution towards the total number of nets from routine distribution and previous mass campaigns are shown by the dashed and dotted blue lines, respectively. The timing of DHS surveys are shown by the red vertical dashed lines for both regular (left) and irregular (right) surveying.

In the case where $\Delta s = \Delta t$, the rate parameter in equation (2) will lead to biased estimates for the decay rate; if surveys are consistently conducted soon after a mass campaign, the empirical age distribution will be biased towards younger nets, such that the fitted rate parameter for the distribution in equation (2) would over-estimate the true decay rate; the converse would hold true when surveys are consistently conducted immediately prior to a mass campaign. Given the frequency of DHS surveys can vary significantly between countries, for countries with fewer surveys, there is an increased risk this may lead to biased estimates in decay rates. For example, if surveys are systematically conducted immediately after a mass distribution campaign, this will lead to over-estimates in the decay rates. Therefore, to attempt to reduce this risk of bias, we use a hierarchical model to pool information over larger spatial scales. To implement this, we use the following positivetuncated normal ($N_{>0}$) prior for the mean lifespan of used nets in region i , with country-specific hyperparameters:

$$\pi(\gamma_i^{-1}) \sim N_{>0}(\gamma_{\zeta(i)}^{-1}, \sigma_{\zeta(i)}^2) \quad (3)$$

For readability, the use or access indicator superscripts have been removed in equation (3) and those that follow. Aside from sub-setting the data for ITNs that were used or provided access, the methodology presented is otherwise identical. Parameter values are allowed to vary; for example, the mean duration of use, $1/\gamma_{\zeta(i)}^u$, and retention time, $1/\gamma_{\zeta(i)}^a$, of ITNs in country $\zeta(i)$ are not assumed to be equal.

The hyperparameters for the country $\zeta(i)$ of region i are then simulated from hyperpriors with universal hyperhyperparameters, as indicated with subscript 0:

$$\pi(\gamma_{\zeta(i)}^{-1}) \sim N_{>0}(\gamma_0^{-1}, \sigma_0^2)$$

$$\pi(\sigma_{\zeta(i)}) \sim N_{>0}(\chi_0, \omega_0^2)$$

We then used a weakly-informative universal hyperhyperprior for the mean retention time with a mean and standard deviation of 24 and 12 months, respectively. Narrower hyperhyperpriors were used for the variance hyperhyperparameters to aid chain convergence during the fitting process:

$$\pi(\gamma_0^{-1}) \sim N_{>0}(24, 12^2)$$

$$\pi(\sigma_0), \pi(\chi_0), \pi(\omega_0) \sim N_{>0}(1, 1)$$

Under our assumption that nets stop being used and are discarded at constant rates within each subnational region, the likelihood, $\mathcal{L}(\gamma_i | a_{il})$, of observing each net with an age a_{il} follows an Exponential (γ_i) distribution. However, as DHS surveys do not record ages of nets greater than 36 months, but report them as being older than this threshold, we therefore right-censor these data, such that:

$$\mathcal{L}(\gamma_i | a_{il}) \sim \begin{cases} f(a_{il}; \gamma_i) & , \text{ if } a_{il} \leq 36 \\ 1 - F(36.5; \gamma_i) & , \text{ otherwise} \end{cases}$$

where $f(\alpha; \gamma)$ and $F(\alpha; \gamma)$ are Exponential density and cumulative distribution functions, respectively:

$$f(\alpha; \gamma) = \gamma e^{-\gamma\alpha}$$

$$F(\alpha; \gamma) = 1 - e^{-\gamma\alpha}$$

To correct for non-random probabilities of households being sampled in DHS surveys, we also conduct the fitting process using pseudo-counts, $M_{i\xi}$, of the number of nets of age α_ξ in region i , where:

$$M_{i\xi} = \left[\sum_{l=1}^{N_{i\xi}} w_{il} \right] \quad (4)$$

given $N_{i\xi}$ is the surveyed number of nets of age $\alpha_{il} = \xi$ in region i , w_{il} is the household survey weight for net l , while $[\cdot]$ denotes a standard rounding function. As DHS surveys have a stratified sampling design each household has a unique probability of being sampled within each region i . To account for these non-random probabilities of households being surveyed, each net l that is recorded has an associated household weighting, w_{il} . The DHS also scales these weightings by a factor of 10^6 ; our notation here of w_{il} corrects for this scaling, and is therefore equal to the reported DHS household weightings divided by 10^6 .

After fitting our model in RStan (v2.32.6, Stan Development Team, 2024), we generated 1,000 samples from each of the four MCMC chains after a burn-in period of 1,000 iterations. This simulated 4,000 draws from the posterior distributions of the mean duration of use and retention time at subnational, $\pi(\gamma_i^{-1}|\alpha)$, national, $\pi(\gamma_{\zeta(i)}^{-1}|\alpha)$, and continental, $\pi(\gamma_0^{-1}|\alpha)$, scales, such that, for example:

$$\pi(\gamma_i^{-1}|\alpha) \propto \pi(\gamma_i^{-1}) \prod_{\xi=0}^{37} \prod_{l=1}^{M_{i\xi}} \mathcal{L}(\gamma_i|\alpha_{il}) \quad (5)$$

Where the observed ages of all sampled nets that were used or provided access is given by $\alpha = [\alpha_1, \dots, \alpha_i, \dots, \alpha_{N_i}]$, such that $[\alpha_i = \alpha_{i,0}, \dots, \alpha_{i,\xi}, \dots, \alpha_{i,37}]$ and $\alpha_{i,\xi} = [\alpha_{i,1}, \dots, \alpha_{i,M_{i\xi}}]$, given that ξ is the observed ages of sampled nets, aside from when $\xi = 37$, which is used to indicate nets older than 36 months.

Estimating the timing of mass campaigns

Prior to estimating the probabilities of use and access subnationally, we first sought to estimate the timings of subnational mass distribution campaigns. We assume that for all subnational regions i within a country $\zeta(i)$, the k^{th} national mass campaign is conducted within a given time interval, $[r_{ik}, s_{ik}] \forall i$ where $\zeta(i) = \zeta$. However, we do not necessarily assume that a national mass campaign occurs in synchrony across all regions.

Using data on the annual numbers of nets delivered to a country, $Z_{\zeta}(t)$, that was compiled by the The Alliance for Malaria Prevention (2024), we applied Nadaraya–Watson kernel regression (Nadaraya, 1964; Watson, 1964) to generate smoothened distributions, $Z_{\zeta}(t)$. This was conducted using the ksmooth function from the R stats package (v4.3.2) for a bandwidth of 12 months due to the original resolution of the data being recorded annually (figure 9.a).

If a net were selected randomly across space and time within a country ζ , we treat $Z_{\zeta}(t)$ as an initial approximation of the probability of the month in which that net was received after normalisation. Under our assumption that mass campaigns are conducted within defined nationwide time intervals, we therefore consider the local minima of the density $Z_{\zeta(i)}(t)$ to also be indicative of the local minima for the probability of a mass campaign occurring in any subnational region i of country $\zeta(i)$ at those times. If the k^{th} mass campaign in region i occurred at time τ_{ik} , we therefore define the range of possible values of τ_{ik} such that $Z_{\zeta(i)}(r_{ik})$ and $Z_{\zeta(i)}(s_{ik})$ are respectively the local minima of $Z_{\zeta(i)}$ immediately preceding and following mass campaign k given $r_{ik} \leq \tau_{ik} < s_{ik}$ and $s_{ik} = r_{i(k+1)}$.

To refine our estimates of when mass campaigns were likely to have occurred subnationally, we initially consider the recorded number of nets received by month in DHS surveys in subnational region i and normalise this to generate another density, $X_i(t)$ (figure 9.b). Given a total N_i nets were recorded across all surveys conducted in region, we generated each density given the age, α_{il} , and survey date, h_{il} , in months for each net $l \in \{1, \dots, N_i\}$:

$$X_i(t) \sim \frac{\sum_{l=1}^{N_i} f(t; \alpha_{il}, h_{il}, w_{il})}{\sum_{l=1}^{N_i} w_{il}} \quad (6)$$

Where:

$$f(t; \alpha, h, w) = \begin{cases} w, & \text{if } t = h - \alpha \\ 0, & \text{otherwise} \end{cases}$$

such that w_{il} is the DHS household survey weight for net l to correct for non-random probabilities of households being surveyed. However, since the density in equation (6) is generated from nets that were observed in households, this density will be systematically biased by younger nets, since there will be a greater probability that an older net would have been discarded before it was recorded in a survey than a younger one. Under our assumption that ITNs are discarded at a constant rate, for every w ITNs surveyed of age α , an expected $w [\exp(\gamma \alpha) - 1]$ ITNs will have been lost between the date they were received and that of the survey. Therefore, to generate an unbiased density, $W_i(t)$ (figure 9.c), we use our previously generated a posteriori mean estimates of mean retention times, $1/\bar{\gamma}_i^a = \mathbb{E}[\pi(1/\gamma_i^a | \alpha)]$, to scale the DHS survey weights by the expected number of ITNs that would have been received for each ITN that was surveyed:

$$W_i(t) \sim \frac{\sum_{l=1}^{N_i} g(t; \alpha_{il}, h_{il}, w_{il}, \bar{\gamma}_i^a)}{\sum_{l=1}^{N_i} w_{il} \exp(\bar{\gamma}_i^a \alpha_{il})}$$

Where:

$$g(t; \alpha, h, w, \gamma) = \begin{cases} w \exp(\gamma \alpha), & \text{if } t = h - \alpha \\ 0, & \text{otherwise} \end{cases}$$

However, to account for months where no nets were recorded to have been distributed, we then generated a composite density with the stepwise normalised density, $Z_{\zeta(i)}(t)$, that was proportional to the national number of nets distributed annually:

$$V_i(t) \propto \begin{cases} Z_{\zeta(i)}(t), & \text{if } W_i(t) = 0 \\ W_i(t), & \text{otherwise} \end{cases} \quad (7)$$

We again conduct Nadaraya–Watson kernel regression on this composite density for each subnational region using the `ksmooth` function in R with a bandwidth of 12 months to generate smoothened densities, $\nu_i(t)$ (figure 9.d). Our rationale for this is two-fold; firstly, the composite density in equation (7) is proportional to the number of annual nets distributed for some months; secondly, as the age distributions of nets for most subnational regions has notable peaks at 12, 24 and 36 months (figure 8), it is probable that responses to this survey question were rounded to the nearest whole year by some participants.

After normalising the density $\nu_i(t)$, such that $\sum_i \nu_i(t) = 1$, this represents our best effort at capturing the probability of the month of distribution if a net is selected at random both in space and time within a subnational region. We then consider the probability density, $\nu_{ik}(t)$, of the distribution date of a net selected randomly conditional on that date being in the interval $[r_{ik}, s_{ik}]$ in which the k^{th} mass distribution campaign occurred:

$$\mathcal{Y}_{ik}(t) \propto \begin{cases} \nu_i(t), & \text{if } r_{ik} \leq t < s_{ik} \\ 0, & \text{otherwise} \end{cases}$$

After discretising this density by month and by normalising such that $\sum_t \mathcal{Y}_{ik}(t) = 1$, we then treat $\mathcal{Y}_{ik}(t)$ as a proxy for the probability of mass campaign k occurring at time τ_{ik} in region i :

$$\mathbb{P}[\tau_{ik} = t] = \mathcal{Y}_{ik}(t)$$

The expected value of the timing of a mass campaign is therefore estimated from:

$$\mathbb{E}[\tau_{ik}] = \sum_{\tau=r_{ik}}^{s_{ik}-1} \tau \mathcal{Y}_{ik}(\tau) \quad (8)$$

To account for uncertainty in the timing of mass campaigns, we then calculate the variance around the timing of each mass campaign:

$$\text{Var}[\tau_{ik}] = \mathbb{E}[\tau_{ik}^2] - \mathbb{E}[\tau_{ik}]^2 \quad (9)$$

Figure 8. Posterior predictive distributions (red) from the hierarchical model for the age of a used net in months if one is sampled randomly over time and across either sub-Saharan Africa (a), or within an exemplar country (b) or subnational region (c).

The shaded regions indicate 95% credible intervals, while the blue lines indicate the normalised empirical densities; these were generated from the pseudo-counts as described in equation (4). The peaks at 12, 24 and 36 months in the empirical densities are believed to be artifacts from the survey data due to individuals rounding reported ITN ages to the nearest whole year.

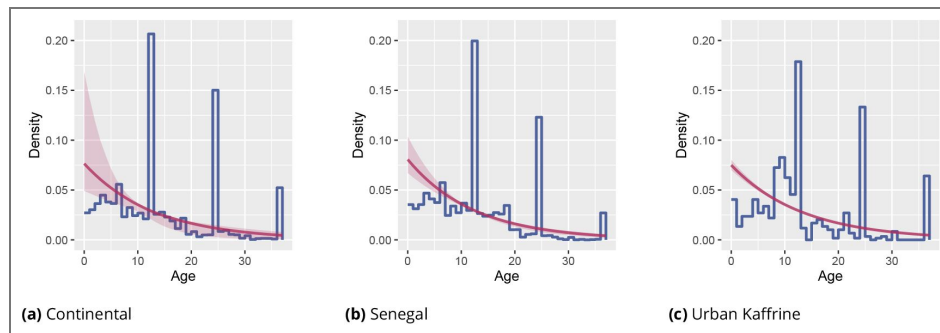
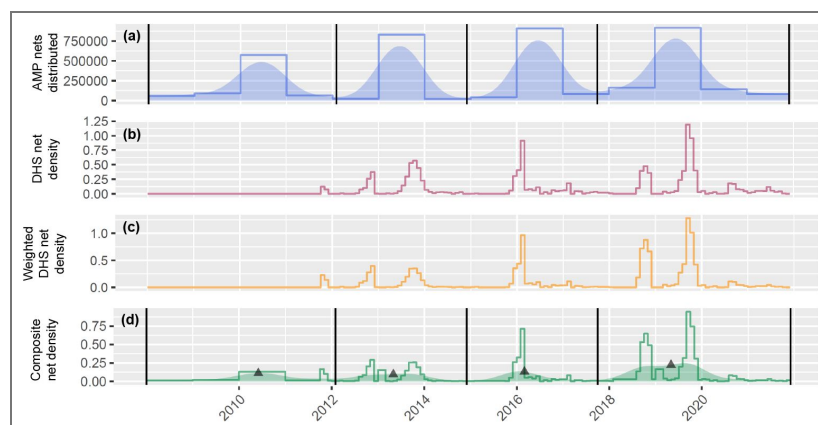


Figure 9. Estimates of mass campaign timings in the subnational Est region in Burkina Faso.

Letting region i be the Est region of Burkina Faso, (a) shows the empirical, $Z_{Z(i)}$, and smoothed, $Z_{\hat{Z}(i)}$, annual numbers of nets delivered nationally are shown by the solid blue line (as reported by the Alliance of Malaria Prevention, AMP) and shaded region. The local minima of $Z_{\hat{Z}(i)}$, which define the midpoints between net delivery dates are shown by the vertical black lines in (a) and (d) and indicate the assumed periods when mass campaigns are allowed to have occurred. The density proportional to the the number of nets received subnationally by month, X_i , is shown in red (b). The weighted version of this density, W_i , to account for older nets being under-represented in DHS surveys is shown in yellow (c). The composite density, V_i , and its smoothed counterpart, $\hat{V}_i$, which are constructed from the stepwise densities in (a) and (c) are shown by the green line and shaded region in (d). The densities, $\hat{V}_{ik}$, are subsetting from $\hat{V}_i$ by the black lines. These are treated as approximations of the probability of a mass campaign occurring in each month, and the expected value for the timing of each k th mass campaign are shown by the black triangles.



Given:

$$\mathbb{E}[\tau_{ik}^2] = \sum_{\tau_{ik}} \tau_{ik}^2 \mathcal{Y}_{ik}(\tau)$$

While this approach represented our best effort at capturing the central tendency of the subnational timings of mass campaigns, the variances calculated from equation (9) are likely to overestimate the degree of uncertainty in campaign timing. For example, for variance around the period of the 2019 mass campaign in Est, Burkina Faso, is far greater than in 2016 (figure 9.d). However, this is potentially driven by a DHS survey begin conducted at the end of 2021, which resulted in a notable number of ITNs being reported as 24 and 36 months old, due to rounding to the nearest whole year. We therefore propose that the peaks in the composite density in figure 9.d are primarily driven by whole year rounding, which has resulted in a large variance in the smoothened density that potentially overinflates the uncertainty in the timing of the 2019 mass campaign.

Due to the potential of over estimating the uncertainty in the timings of mass campaigns due to whole year rounding in survey responses, we rank the variances calculated from equation (9) for all mass campaigns in all regions. We then split these into tertiles of equal size based on the variances calculated from equation (9) and assign standard deviations for the timings of mass campaign k in region i , $\sigma_{\tau_{ik}}$ of one, two and three to represent mass campaigns where there is low, medium and high uncertainty in the timing of mass campaigns. We then approximate the uncertainty around our central estimates with a truncated-normal (TN) density with upper and lower bounds set at three standard deviations from the expected value:

$$\pi(\tau_{ik} | X_i, Z_{\zeta(i)}) \stackrel{\text{approx}}{\sim} \text{TN}(\bar{\tau}_{ik}, \sigma_{\tau_{ik}}^2, a_{ik}, b_{ik}) \quad (10)$$

Where:

$$a_{ik} = \bar{\tau}_{ik} - 3\sigma_{\tau_{ik}}^2$$

$$b_{ik} = \bar{\tau}_{ik} + 3\sigma_{\tau_{ik}}^2$$

and $\bar{\tau}_{ik} = \mathbb{E}[\tau_{ik}]$ from equation (8). This gives a range of proposed dates for the area which is desirable as campaigns are likely to be spread out over a period given the geographical scale of the area and the considerable logistics involved. The proposed timings of mass campaigns were verified where possible though discussion with interested parties nationally and internationally, though further work to record the subnational timings of mass campaigns is encouraged.

Estimating historical use and access and refining retention times

The model presented in this section describes how we estimated historical trends in use. Aside from fitting the model to the number of individuals who had access to nets, a_{ij} , instead of those who used nets, u_{ij} , in region i at time t_j , the methodology is otherwise identical to how we estimated historical trends in access.

We firstly define $\theta_i = \{\psi_i, \beta_i^0, \beta_i^t, q_i^0, \lambda_i\}$ as a vector of unknown parameters in our model; the elements of which we will later define. To construct estimates of the probability p_{ij} that an individual uses a net in region i at time $t_j \ni t$, given that u_{ij} of n_{ij} surveyed individuals used a net at time t_j in admin i , we implement the following likelihood:

$$\pi(u_{ij} | \theta_i, p_{ij}) \sim \text{Binomial}(n_{ij}, p_{ij})$$

Assuming individuals only sleep under a single net, the probability that an individual will use an ITN, p_{ij} , at time t_j will be equal to the probability of using a continuously-distributed ITN, d_{ij} before the timing of the first mass campaign, τ_{i1} , and the sum of d_{ij} and the probability of using a campaign-sourced ITN, c_{ij} once mass campaigns have been introduced:

We then assume there exists an upper threshold in the probability of use, p_{ij}^0 , that is achieved immediately following a mass campaign, which will exhibit logistic growth over time and tend towards a saturation threshold, ψ_i :

$$p_{ij}^0 = \psi_i \text{logit}^{-1} (\beta_i^0 + \beta_i^t t_j)$$

Since little is known of the parameters ψ_i , β_i^0 and β_i^t , we assign a non-informative Jeffrey's prior to k_i and weakly informative priors to β_i^0 and β_i^t , and impose a restriction that $\beta_i^t > 0$ to represent our prior belief that the probability of use following mass campaigns will strictly increase over time:

$$\pi(\psi_i) \sim \text{Beta}(0.5, 0.5)$$

$$\pi(\hat{\beta}_i^0) \sim N(0, 0.1)$$

$$\pi(\hat{\beta}_i^t) \sim N_{>0}\left(\frac{\sigma_t}{\bar{t}}, 0.1\right)$$

Where:

$$\beta_i^0 = \hat{\beta}_i^0 - \frac{\hat{\beta}_i^t}{\sigma_t} \bar{t}$$

$$\beta_i^t = \frac{\hat{\beta}_i^t}{\sigma_t}$$

and $\hat{t}_j \ni \hat{t}$ is a standardised survey time point such that $\mathbb{E}[\hat{t}] = 0$ and $[\hat{t}] = 1$, given $\hat{t}_j = (t_j - \bar{t}) / \sigma_t$, $\mathbb{E}[t] = \bar{t}$ and $[t] = \sigma_t^2$.

Assuming the odds $q_i^0 / (1 - q_i^0)$ of a campaign-sourced ITN to a continuously-distributed ITN being used is constant for all points in time immediately following a mass campaign, the probabilities here for using campaign-sourced and non-campaign-sourced ITNs are respectively:

$$c_{ij}^0 = q_i^0 p_{ij}^0$$

$$d_{ij} = (1 - q_i^0) p_{ij}^0$$

Where $c_{ij}^0 = c_{ij}$ when the number of months following a mass campaign, $m_{ij} = 0$. The probability of a net being sourced from a campaign, given that a net was used, is therefore given by q_i^0 when $m_{ij} = 0$, for which we assume a non-informative Jeffrey's prior:

$$\pi(q_i^0) \sim \text{Beta}(0.5, 0.5)$$

We now note we have implicitly assumed that continually-distributed nets are replenished in the population to ensure their probability of use grows logistically over time, since:

$$d_{ij} = (1 - q_i^0) \psi_i \text{logit}^{-1} (\beta_i^0 + \beta_i^t t_j) \quad (12)$$

However, we would expect the conditional probability of a used net to be sourced from a mass campaign q_{ij} to decrease over time following the last mass campaign such that $c_{ij} = q_{ij} p_{ij}$. Therefore, under our previous assumption that the probability of an individual net being used decays exponentially over time, it follows that:

$$c_{ij} = c_{ij}^0 \exp(-\lambda_i m_{ij}) = q_i^0 \psi_i \exp(-\lambda_i m_{ij}) \text{logit}^{-1} (\beta_i^0 + \beta_i^t t_j) \quad (13)$$

Where our previous a posteriori estimates of the mean duration of use from [equation \(5\)](#) are used to inform strongly informative priors:

$$\pi(\lambda_i^{-1}) \sim N(\mu_{\lambda_i}, \sigma_{\lambda_i}^2)$$

Given:

$$\mathbb{E} \left[\pi \left(\gamma_i^{-1} \middle| \alpha \right) \right] = \mu_{\lambda i}$$

$$\text{Var} \left[\pi \left(\gamma_i^{-1} \middle| \alpha \right) \right] = \sigma_{\lambda i}^2$$

We herein assume an individual that has used, or has access to, a net has the same probability of receiving an ITN as an individual who has not used, or does not have access to, an ITN. Provided the number distributed through continuous channels is sufficiently small, it can be shown that in region i , the mean duration of use (or retention time in the context of access) under the assumption of random allocation of ITNs, Λ_i , is approximated by:

$$\Lambda_i^{-1} \approx \frac{p_{ij}^0}{c_{ij}^0 \lambda_i} = q_i^0 \lambda_i^{-1}$$

The time-varying conditional probability of using a campaign-sourced ITN can in turn be derived from [equations \(12\)](#) and [\(13\)](#):

$$q_{ij} = \frac{c_{ij}}{p_{ij}} = \frac{q_i^0 \exp(-\lambda_i m_{ij})}{q_i^0 \exp(-\lambda_i m_{ij}) + (1 - q_i^0)}$$

Given the source of ITNs is recorded in most DHS surveys, we can describe the likelihood of the number of used nets sourced from mass campaigns, y_{ij} , with the following binomial distribution:

$$\pi(y_{ij} | \theta_{ij}) \sim \text{Binomial}(u_{ij} | q_{ij})$$

Letting $\bar{y}_i$ be the observed proportion of used nets in region i that originated from campaigns, for each net l where the source was not recorded, we probabilistically impute whether they originated from campaigns:

$$y'_{il} \sim \text{Bernoulli}(\bar{y}_i)$$

Now, if the timing of the k^{th} campaign that most recently preceded a survey point at time t_j was known ($\phi_{ij} \in \tau_{ik}$), then the months since the last campaign could be directly inferred from:

$$m_{ij} = t_j - \phi_{ij} \quad (14)$$

However, while we do not know the exact timings of when campaigns were conducted, we captured our uncertainty around our estimates of when these occurred through [equation \(10\)](#). Before we can probabilistically represent the months since the last mass campaign occurred, we need to probabilistically estimate the timing of the last campaign relative to each time point t_j for each region i . Before proceeding, we note that the supports from [equation \(10\)](#) for different campaigns in the same region i are non-overlapping. For times $t_j < a_{i1}$ that precede the support $[a_{i1}, b_{i1}]$ of the distribution in [equation \(10\)](#) for the first campaign when $k = 1$, we therefore assume a zero probability of there being a preceding campaign so that the probability of using any net here is equal to the probability of using a continuously-distributed net, $p_{ij} = d_{ij}$, as per [equation \(11\)](#). We can also represent this as a unit point mass ($\delta_{-\infty}$) for a campaign occurring at $t = -\infty$, since $\lim_{\phi_{ij} \rightarrow -\infty} m_{ij} \rightarrow \infty$ from [equation \(14\)](#) and $\lim_{m_{ij} \rightarrow \infty} c_{ij} = 0$ from [equation \(13\)](#). For times within the support of the first campaign, there will be a non-zero probability that the first campaign was the most recent; the remaining probability would be that there was no preceding campaign. For times within the support of any other campaign, there will be a non-zero probability that campaign was the most recent, while the remaining probability would be that the earlier campaign was the most recent. Finally, when there is a zero probability of a campaign having occurred at time t_j in region i (i.e. times that fall outwith the supports of distributions from [equation \(10\)](#)), then provided the first campaign has elapsed, there is an assumed 100% probability of the campaign with the most recent central estimate being the most recent. We can therefore probabilistically describe the timing of the last campaign relative to time t_j in region i as follows:

$$\pi(\phi_{ij} | X_i, Z_{\zeta(i)}, t_j) \sim \begin{cases} \delta_{-\infty} & , \text{if } t_j < a_{i1} \\ \rho(s \in [a_{i1}, t_j]) + \delta_{-\infty} \int_{t_j}^{b_{i1}} \rho(s) ds & , \text{if } a_{i1} \leq t_j < b_{i1} \\ \rho(s \in [a_{i\kappa}, t_j]) + \rho(s \in [a_{i(\kappa-1)}, b_{i(\kappa-1)}]) \int_{t_j}^{b_{i\kappa}} \rho(s) ds & , \text{if } a_{i\kappa} \leq t_j < b_{i\kappa} \text{ and } \kappa \geq 2 \\ \rho(s \in [a_{i\kappa}, b_{i\kappa}]) & , \text{otherwise} \end{cases} \quad (15)$$

Given, from equation (10):

$$\rho(\tau_{ik}) = \pi(\tau_{ik} | X_i, Z_{\zeta(i)})$$

where κ denotes the latest campaign with a non-zero probability of being the most recent relative to time t_j such that $b_{i\kappa} - t_j = \min_k(b_{ik} - t_j)$. Ultimately, this enables the months since the last campaign to be probabilistically described:

$$\pi(m_{ij} | X_i, Z_{\zeta(i)}, t_j) = t_j - \pi(\phi_{ij} | X_i, Z_{\zeta(i)}, t_j) \quad (16)$$

This concludes the description of the binomial model to estimate the probability of net use over time. A limitation of this model is the assumption that the probability p_{ij} of using a net in administrative-unit i at time t_j is constant for all individuals. Examination of the raw data (for example, figure 1.A) indicates substantial variability in the recorded proportions of use and access in adjacent months. This may reflect substantial intraregional variability, and there are likely other covariates that our model does not account for that will create additional variability in the probability of use within an administrative-one unit at any given point in time. Therefore, to account for overdispersion, we utilised an additional beta-binomial likelihood function for the observed number of individuals who used a net:

$$\pi(u_{ij} | \theta_{ij}) \sim \text{Beta-Binomial}(n_{ij}, \alpha_{0i} p_{ij}, \alpha_{0i}(1 - p_{ij}))$$

Under the beta-binomial model, posterior estimates for p_{ij} are derived using the same approach as for the binomial model. We also generate posterior estimates for the overdispersion hyperparameter, α_{0i} , given weakly informative hyperpriors, $\pi(\alpha_{0i}) \sim \text{Exponential}(10^{-3})$.

It should be noted that higher values of α_{0i} represent lower levels of overdispersion, and in the limit $\alpha_{0i} \rightarrow \infty$, the beta-binomial model decomposes to the binomial model. Furthermore, the two models are complimentary since both models will yield the same mean estimates of the probability of use p_{ij} . Moreover, while the binomial model can be interpreted as yielding subnational estimates of the population mean probability of use at each time point, the beta-binomial model will yield estimates of the probability of use for an individual selected at random within subnational populations over the same time series. This provides a measure of equity in use and access of nets within a region which will have important policy implications for distribution strategies.

Finally, while we fitted the above models directly to observed DHS data, hereafter we accounted for the potential of self-reported ITN use through DHS surveys to be greater than objectively measured use. A meta-analysis by Krezanoski et al. (2018) estimated self-reported ITN use is typically 8% (95% CI: 3, 13) higher than the true value, or a 13.6% relative overestimation. As the majority of regions were estimated to have levels of use greater than 90% immediately following a campaign by 2024 under our model, and since use following mass campaigns is estimated to have meaningfully increased over 2018-2024 for most regions, we take a conservative approach in accounting for over-reporting and purposefully treat the absolute values estimated by Krezanoski et al. (2018) as relative values to the reported use. The scaled probability of use, $\tilde{p}_{ij}$ is herein assumed to be:

$$\tilde{p}_{ij} = (1 - \epsilon_i) p_{ij}$$

Where:

$$\epsilon_i \sim N_{>0}\left(0.08, \left(\frac{0.05}{1.96}\right)^2\right) \quad (17)$$

This under-reporting assumption is applied for both campaign-distributed and continuously-distributed ITNs. While we do not assume a similar systematic discrepancy between surveyed access and true access, the under-reporting assumption applied to use is accounted for in our calculations of use given access.

Projections of cases averted under different net distribution strategies

All scenarios assume ongoing continuous distribution of ITNs and are parameterised from estimates at the end of 2024 from the earlier statistical model (figure 2.A) and are calibrated against annual estimates of *P. falciparum* prevalence in 2-10 year old children ($PfPR_{2-10}$) estimated by the Malaria Atlas Project (2024) (figure 2.B). Forecasted increases in pyrethroid resistance are accounted for, while other interventions are assumed to be maintained in each region under different scenario forecasts. The transmission dynamics model, which incorporates typical seasonal patterns in each region, is broadly able to reproduce the observed changes in $PfPR_{6-59mo}$, as estimated by DHS surveys, providing some confidence in the resulting epidemiological predictions. However, while the model captures typical seasonal dynamics, it does not account for inter-annual variation or specific weather events. These factors may have contributed to larger deviations from DHS $PfPR_{6-59mo}$ point estimates at certain times; estimates that themselves often have wide 95% credible intervals.

Future projections were conducted using *malariasimulation* (v1.6.0) (Charles et al., 2024), which encodes an individual-based *P. falciparum* transmission dynamics model. The core structure and parameterisation of the model are described in Griffin et al. (2010, Griffin et al. 2014) and are detailed in full at <https://mrc-ide.github.io/malariasimulation/index.html>. Aside from ITN-interventions, transmission dynamics models were calibrated for rural and urban settings in each administrative-one unit using the *site* (v0.2.2) (<https://mrc-ide.github.io/site/>) (Winskill, 2024) package, which has compiled location specific data from multiple sources for 2000 to 2024. *site* characterises local human population structure from WorldPop population estimates (Tatem, 2017), and vector abundance and distribution by species from Malaria Atlas Project (2024) estimates. *site* has also compiled treatment rates and coverage of non-ITN interventions, including indoor residual spraying of insecticide (IRS), seasonal malaria chemoprevention (SMC), drug treatment and vaccination from historical estimates by Malaria Atlas Project (2024), DHS STATcompiler (2022), ACCESS-SMC (2022), SMC Alliance (2022), and the Malaria Vaccine Implementation Programme (Hamel, 2021). Model seasonality is characterised (Winskill, 2024) by fitting Fourier series to CHIRPS (Funk et al., 2014) daily rainfall data for each administrative-one unit. A linear relationship following a 30-day delay between the rainfall Fourier series and larval carrying capacity is assumed in each location following White et al. (2011). Administrative-one level estimates of pyrethroid resistance have been estimated using spatio-temporally distributed bioassay mortality data (Winskill, 2022).

After accounting for population structure and non-ITN interventions, we calibrated the baseline transmission intensity (i.e. prior to the introduction of control interventions) using the *Cali* (v1.0.8) (<https://mrc-ide.github.io/cali/>) package. *Cali* utilises an optimisation algorithm to minimise the sum of absolute differences between a desired prevalence series and one predicted by a model. For this process we fitted historical model estimates of $PfPR_{2-10}$ to annual $PfPR_{2-10}$ estimates from the Malaria Atlas Project (2024) over 2005 - 2024 (figure 1.B). As a means of validation, we compared our model estimates of $PfPR_{6-59mo}$ to estimates from DHS surveys (figure 1.B). We have also quantified the uncertainty in the estimate of the true $PfPR_{6-59mo}$ ($\mathcal{P}$) attributable to the number of children tested. Using a conjugate Beta-Binomial model with a non-informative Beta(0.5, 0.5) prior, the posterior distribution of $\mathcal{P}$ is:

$$\mathcal{P} \sim \text{Beta}(x^+ + 0.5, x^- + 0.5)$$

Where $x^+ = \sum_{i=1}^{n^+} w_i$, and $x^- = \sum_{i=1}^{n^-} w_i$, given n^+ and n^- are the number of 6-59 month olds who tested positive and negative, respectively, and w_i is the scaled sample weight for child i such that $x^+ + x^- = n^+ + n^-$.

The insecticidal activity of an ITN is assumed to decay at a constant rate, γ_N . Conditional on a feeding attempt on a human using an ITN of age ξ , the probability that a mosquito successfully feeds (s_N), or is repelled (r_N) or killed (d_N) by that ITN is described by:

$$r_N = (r_{N0} - r_{NM}) \exp(-\xi\gamma_N) + r_{NM} \quad (18)$$

$$d_N = d_{N0} \exp(-\xi\gamma_N) \quad (19)$$

$$s_N = 1 - r_N - d_N \quad (20)$$

Where r_{N0} and d_{N0} are the respective probabilities of repellency and induced mortality of a new ITN, while r_{NM} is the long-term probability of repellency from the physical barrier alone. Following a systematic review by Nash et al. (2021) that characterised the relationship between the level of resistance in wild mosquito populations, using the probability of bioassay survival as a proxy, and the probability of surviving 24 hours following exposure to a pyrethroid-only ITN, Sherrard-Smith et al. (2022) estimated the parameters in equations (18) - (20) for a range of pyrethroid-resistance levels. This was conducted for both pyrethroid-only and pyrethroid-PBO ITNs from a suite of paired EHT trials. This work has since been updated by Churcher et al. (2024) in the context of pyrethroidchlorfenapyr ITNs and the models are broadly able to recreate the observed epidemiological benefit of different classes of ITNs evaluated in two cluster randomised control trials. Given estimated levels of pyrethroid resistance for each location over time, the probabilities of repellency and induced mortality for pyrethroid-only, pyrethroid-PBO and pyrethroid-chlorfenapyr nets were parameterised for our transmission dynamics model accordingly.

For each subnational region, 100 realisations were simulated for all ITN distribution strategies considered (biennial and triennial mass campaigns with pyrethroid-only, -PBO and -chlorfenapyr ITNs) for a population of 100,000 individuals. We initialise all transmission models in January 2000 for each location, and assume continuous distribution over 2000-2007 at the same level as our historical estimates for January 2008. We then utilise our historical estimates of use from 2008 until our central estimates of the time of the last campaign, τ_L , before 2025. Given a mass campaign interval of $\Delta\tau$, the timing of the first campaign of future projections, τ_F , is then drawn randomly from the interval $\tau_L + \Delta\tau \pm 6$ months. Subsequent mass campaigns occur after every $\Delta\tau$ interval thereafter. Pyrethroid-only ITNs are assumed to be distributed prior to 2025 through both continuous and mass campaign channels. For pyrethroid-PBO and pyrethroid-chlorfenapyr scenarios, all ITNs distributed from τ_F onwards are assumed to be of the new ITN type, again in both continuous and mass campaign contexts. Continued logistic growth in the number of ITNs distributed through continuous channels or in total use following a mass campaign is no longer assumed beyond 2022, respectively modifying equations (12) and (13) for the probabilities of using an ITN from continuous and mass campaign channels accordingly:

$$d_{ij} = (1 - q_i^0) f_i(t_j) \quad (21)$$

$$c_{ij} = q_i^0 \exp(-\lambda_i m_{ij}) f_i(t_j) \quad (22)$$

Where:

$$f_i(t) = \begin{cases} g_i(\text{January-2008}) & , \text{ if } t < \text{January-2008} \\ g_i(\text{December-2024}) & , \text{ if } t > \text{December-2024} \\ g_i(t) & , \text{ otherwise} \end{cases} \quad (23)$$

$$g_i(t) = \psi_i \text{logit}^{-1}(\beta_i^0 + \beta_i^t t) \quad (24)$$

For each realisation, values of q_i^0 , λ_i , ψ_i , β_i^0 and β_i^t in equations (21) - (24) are drawn from the joint posterior distribution from our regression model of historical use, which was fitted using RStan (v2.32.6, Stan Development Team, 2024), while the degree of over-reporting of use is simulated from equation (17). Every 100 realisations for each scenario is simulated over 2000 to 2030 with the mean cases averted over the final nine years recorded. We then restrict this to a

period of six years following the first mass campaign under future projections allow direct comparisons of cases averted between two triennial or three biennial campaigns; median time windows are used for future projections in the absence of mass campaigns. The maximum use estimated to be achieved by a campaign, had one been conducted at the end of 2024, is assumed to be maintained under future projections for each region. Similarly, the level of use of continuously-distributed ITNs estimated at the end of 2024 for each region is assumed to be maintained. It is assumed that for a particular subnational region, the use achieved after a biennial campaign is the same as that under triennial campaigns. We generate estimates of clinical cases under different distribution strategies by simulating campaign intervals of two and three years, in addition to the cessation of mass campaigns entirely, over the six-year projections. When switching from triennial to biennial campaigns, the model assumes those already with access are not excluded from mass campaigns, and accounts for diminishing returns in access as nets-per-capita increases (Bertozzi-Villa et al., 2021 [↗](#)). It also assumes that use given access changes in the same manner to what was observed in triennial campaigns. All ITNs are assumed to be of the same type within a subnational region, and remain the same over subsequent mass campaigns. Routine distribution of ITNs, IRS, SMC and the level of drug treatment is assumed to continue as normal at the observed level. Posterior predictive estimates of cases averted are calculated from the additional cases averted over a base-line scenario where no future ITNs are distributed beyond our historical estimates (and those with nets loose them over time).

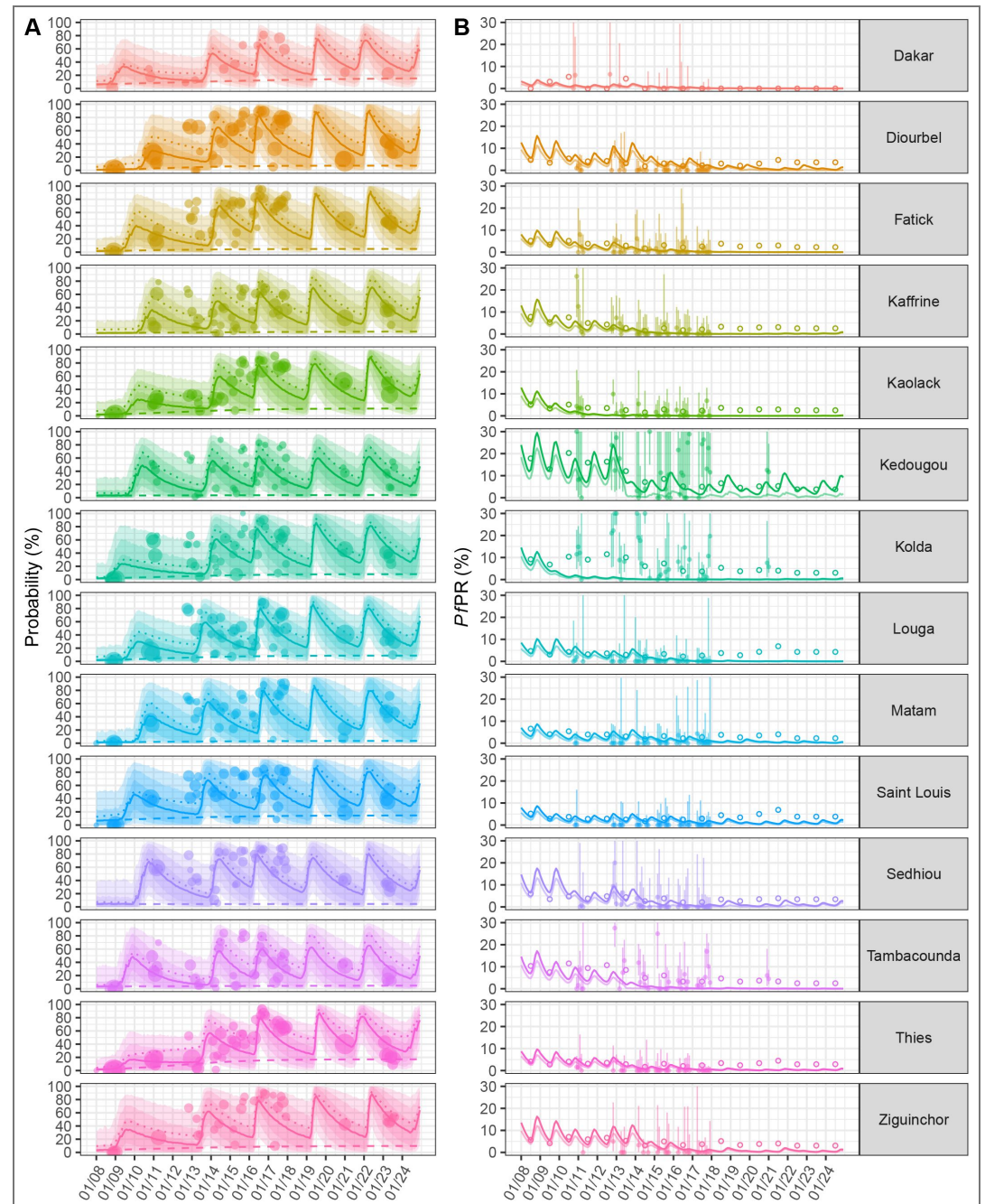


Figure 2—figure supplement 1. Use, access and *Pf* PR over time in rural Senegal. Model estimates in a lower transmission context in Senegal are shown. All features remain unchanged from figure 1. Note, in B, some credible intervals extend beyond the y-axis limits, which have been restricted to 30% to aid visibility.

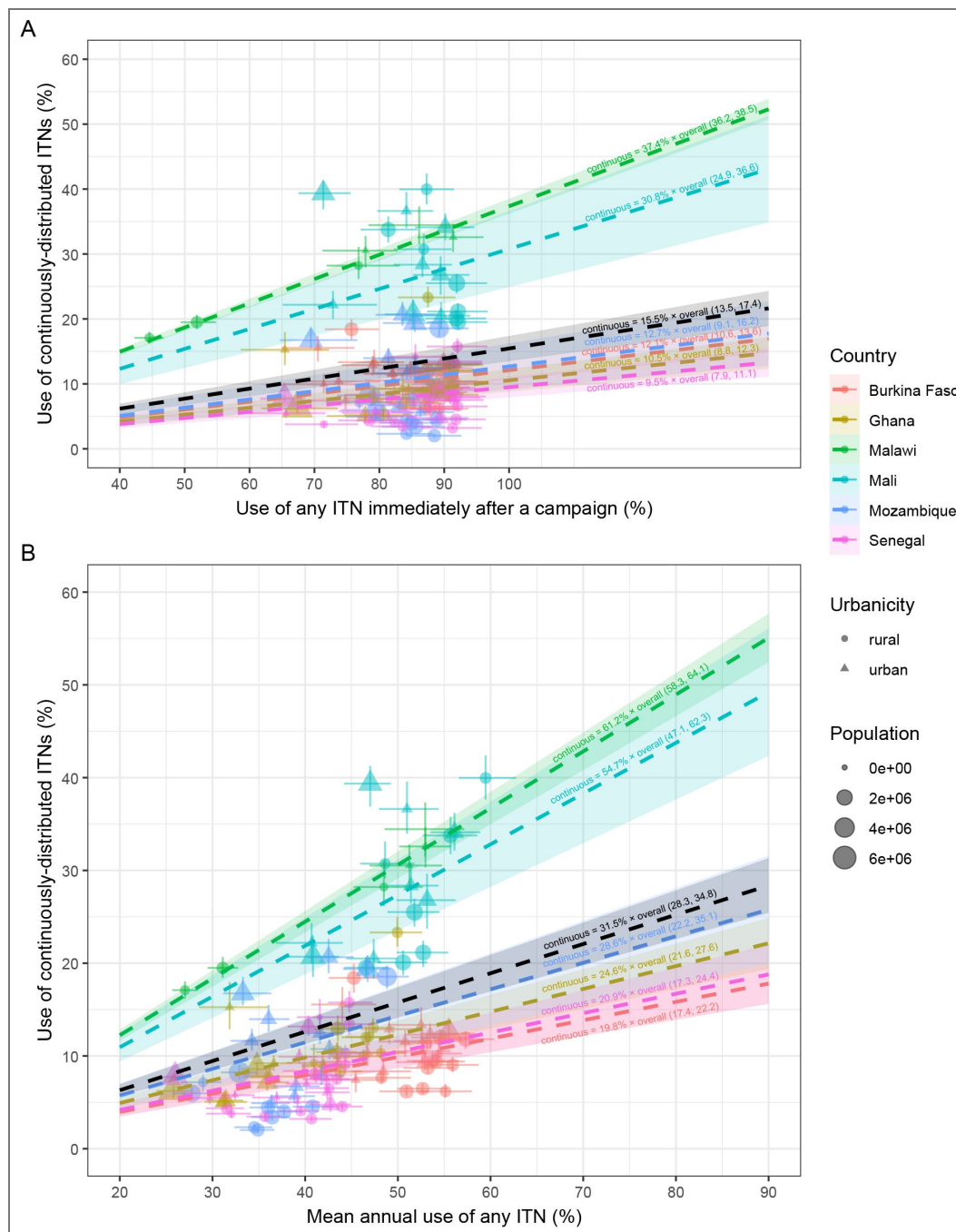


Figure 2—figure supplement 2. Use of ITNs from continuous channels.

For each subnational region, coloured points indicate median estimates of the mean use of any ITN, and of those sourced from continuous channels, immediately following a mass campaign (A) and over the subsequent three years (B) with associated 95% credible intervals. For each country, the proportion of ITNs that were used which were sourced from continuous channels immediately following a campaign (A) and over the subsequent three years (B) can be estimated from the slopes of the coloured population-weighted linear regression lines, which assume a zero intercept. Estimates over all countries are shown by the population-weighted linear regression line in black.

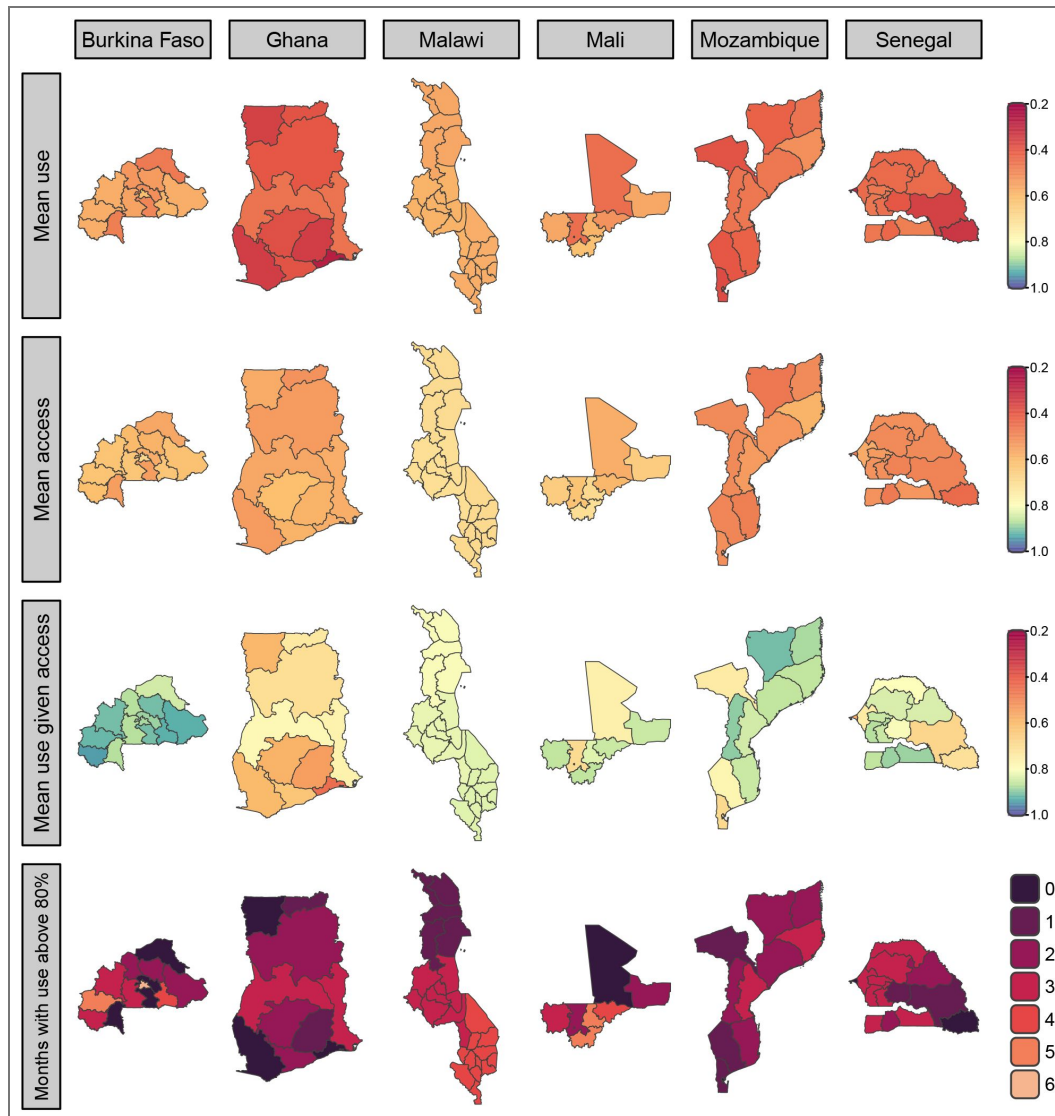


Figure 3—figure supplement 1. Mean urban use and access with three-year campaigns.

Central estimates of mean overall proportion of people using an ITN the previous night, access to an ITN, and use given access for three-year mass campaign intervals are shown in the top three rows. The number of months following a mass campaign where overall ITN use exceeds 80% are shown in the bottom row.

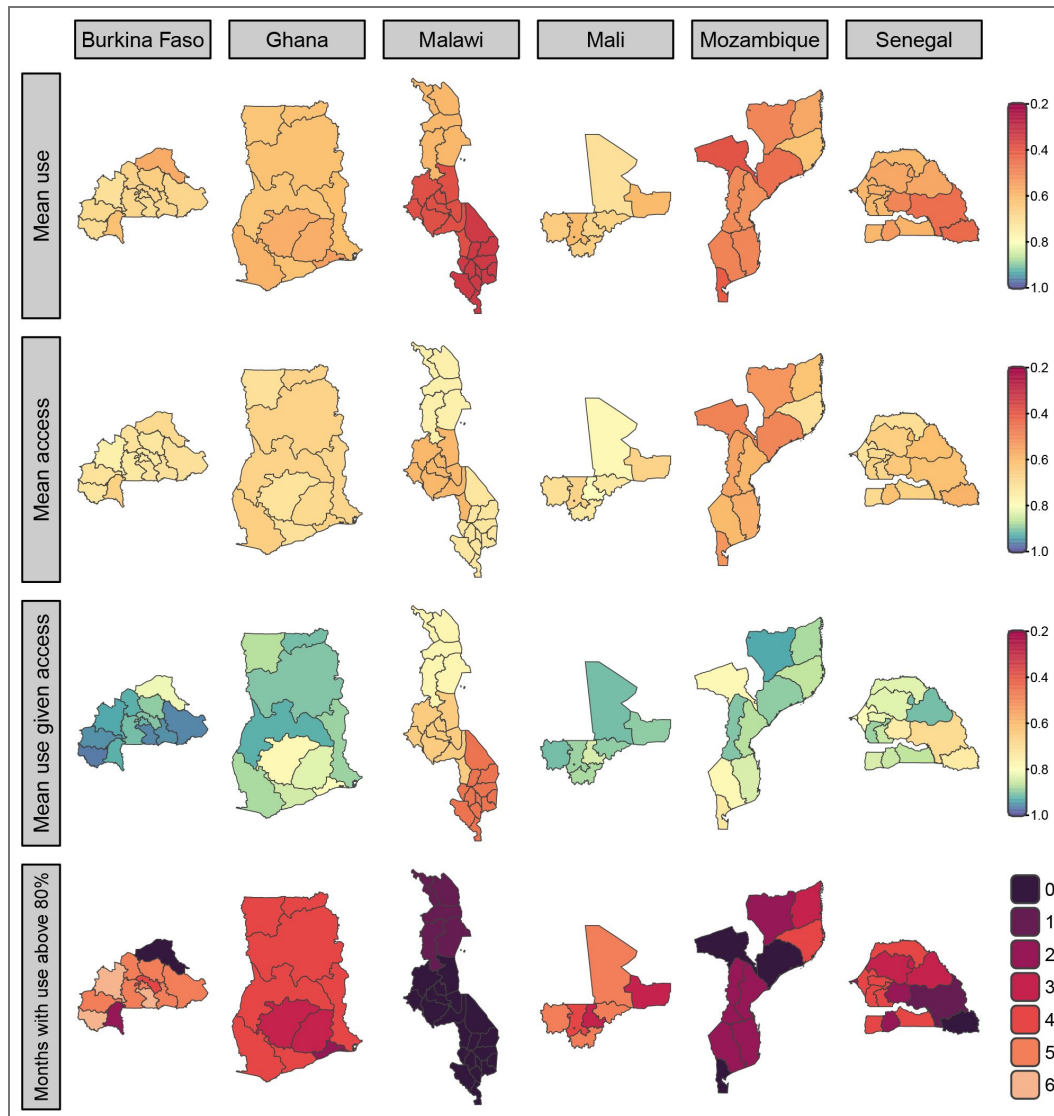


Figure 3—figure supplement 2. Mean rural use and access with two-year campaigns.

Central estimates of mean overall subnational ITN use, access and use given access for two-year mass campaign intervals in rural areas are shown in the top three rows. The number of months following a mass campaign where overall ITN use exceeds 80% are shown in the bottom row.

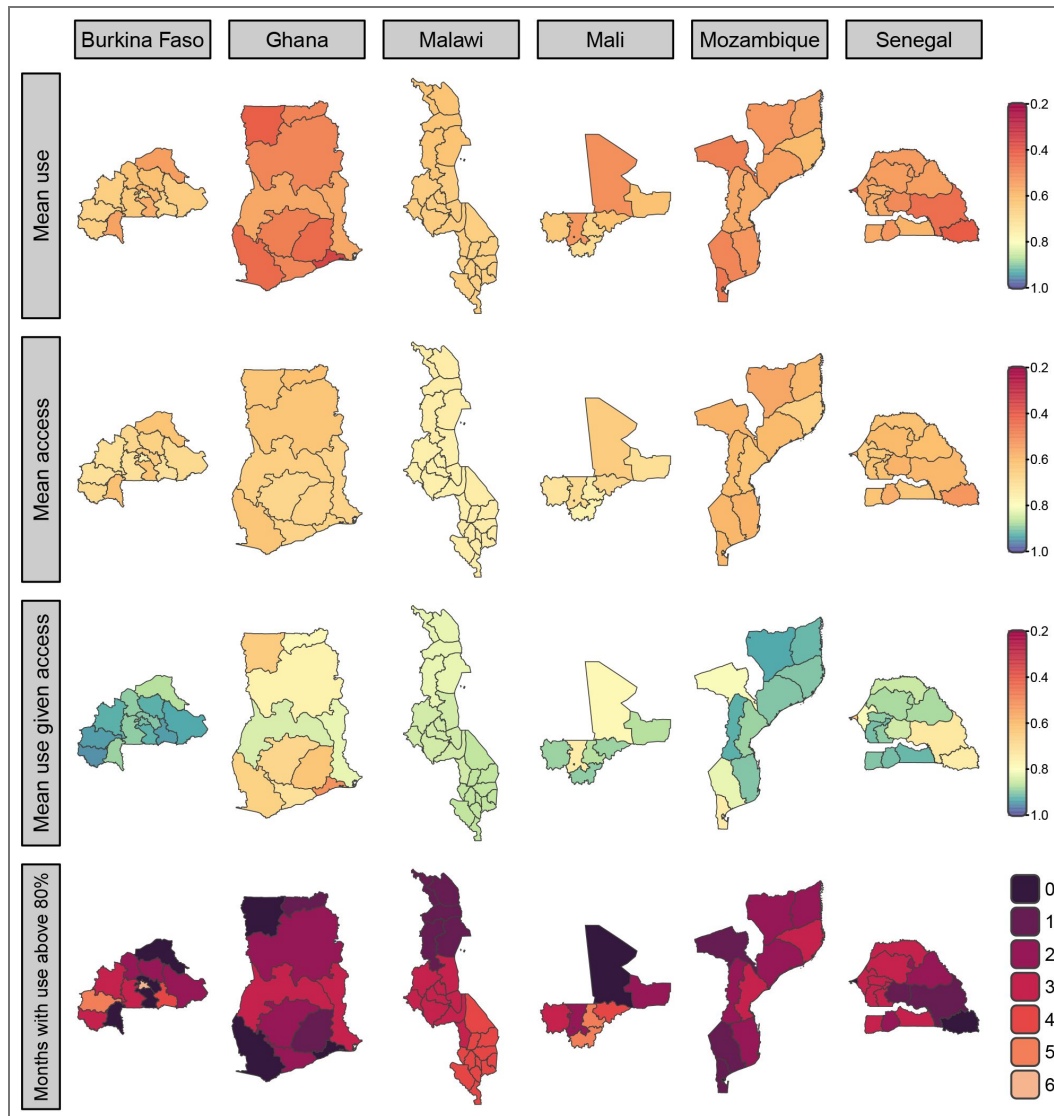


Figure 3—figure supplement 3. Mean urban use and access with two-year campaigns.

Central estimates of mean overall subnational ITN use, access and use given access for two-year mass campaign intervals in urban areas are shown in the top three rows. The number of months following a mass campaign where overall ITN use exceeds 80% are shown in the bottom row.

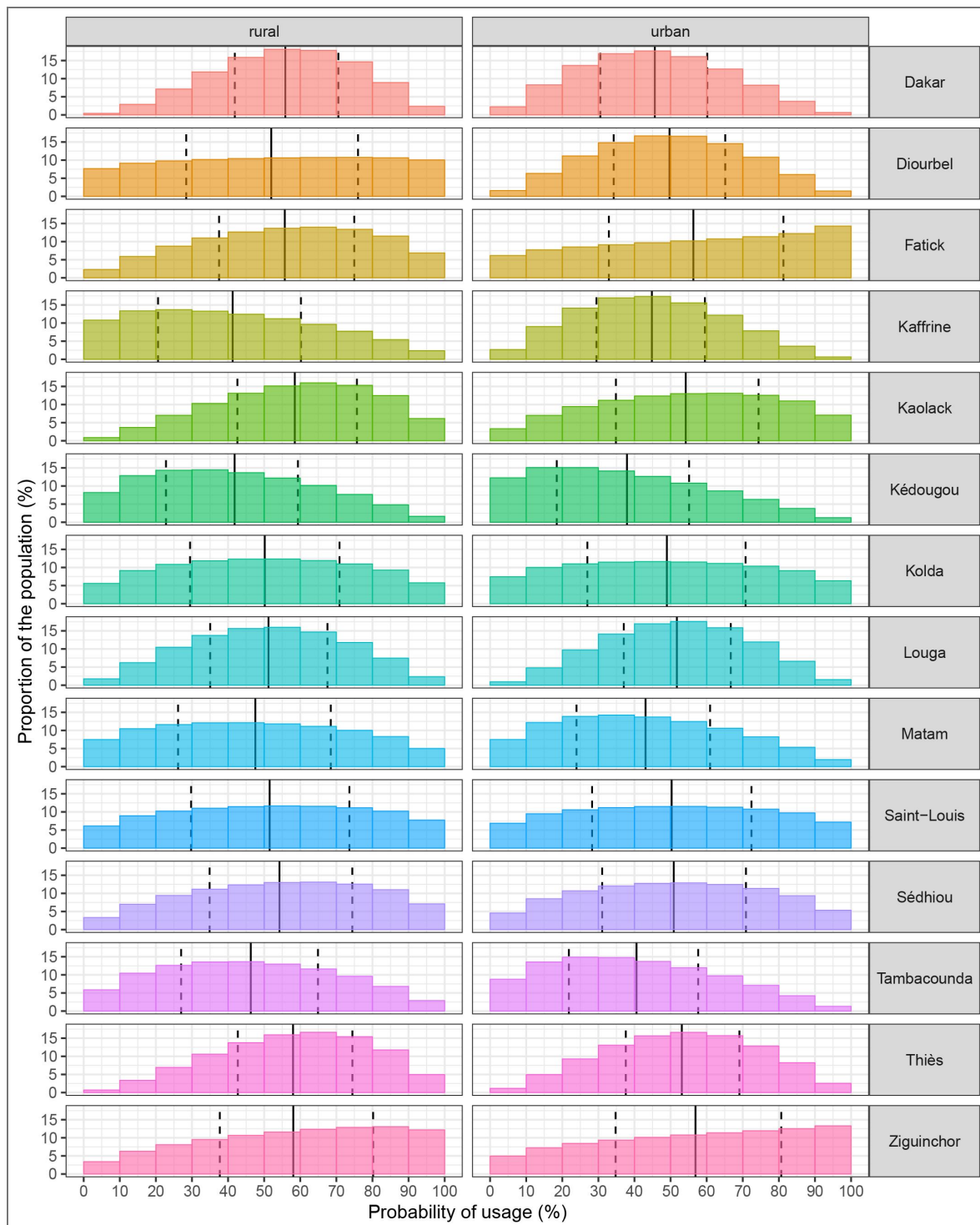


Figure 4—figure supplement 1. Subnational equity of use in Senegal.

All features remain unchanged from [figure 4](#).

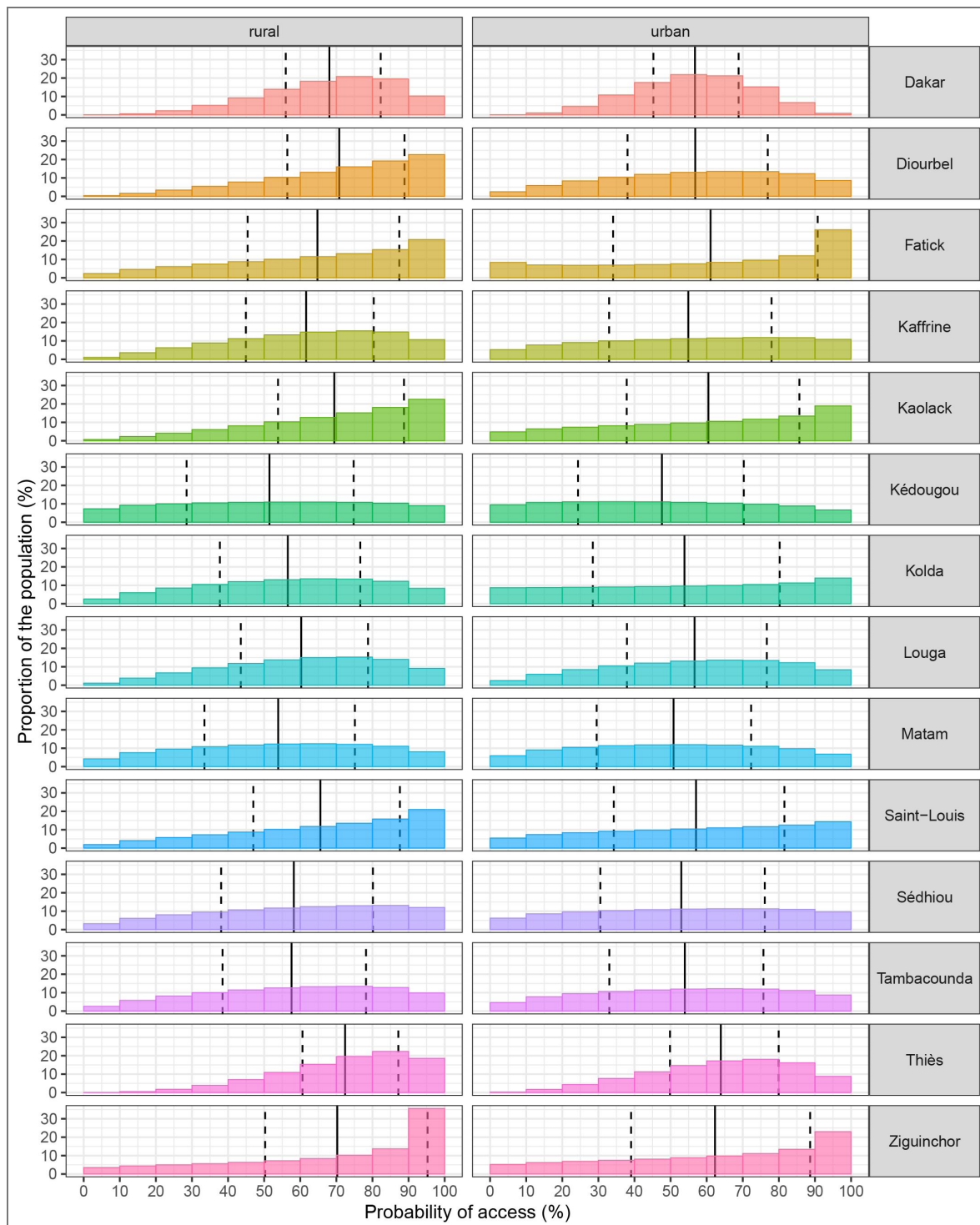


Figure 4—figure supplement 2. Subnational equity of access in Senegal.

All features remain unchanged from [figure 4](#).

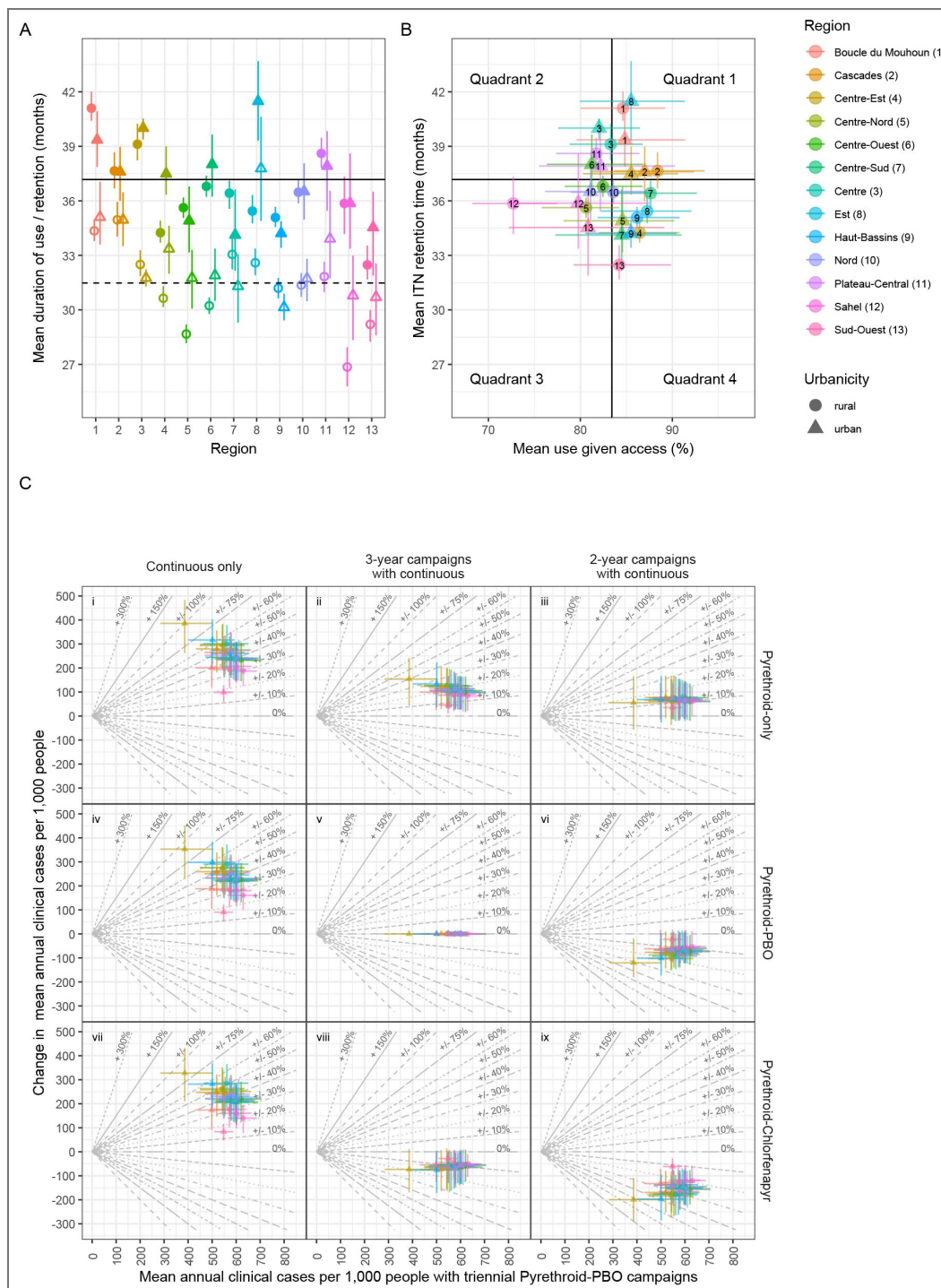


Figure 5—figure supplement 1. Retention time, use given access and changes in cases for Burkina Faso.

All features remain unchanged from [figure 5](#).

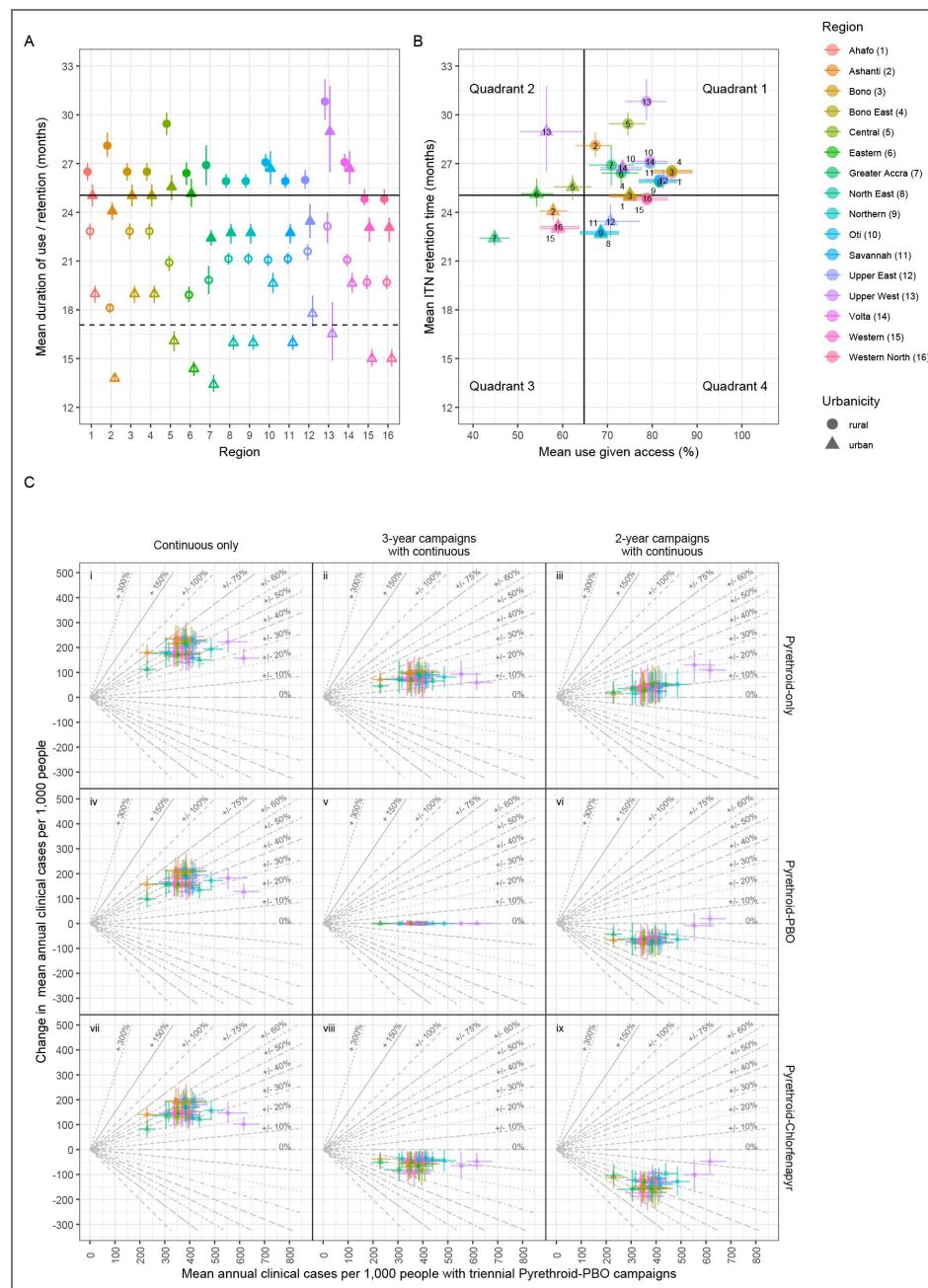


Figure 5—figure supplement 2. Retention time, use given access, and changes in cases for Ghana.

All features remain unchanged from figure 5. In 2019, several administrative regions in Ghana were subdivided: Brong-Ahafo became Bono, Bono East, and Ahafo; the Northern Region was split into Northern, Savannah, and North East; and Volta was divided into Volta and Oti. Retrospective analyses (A–B) were conducted using the pre-2019 regional boundaries, while future projections (C) used the post-2019 subdivisions. In all subnational regions, ITN retention times and use given access were estimated to be higher in rural than in urban settings.

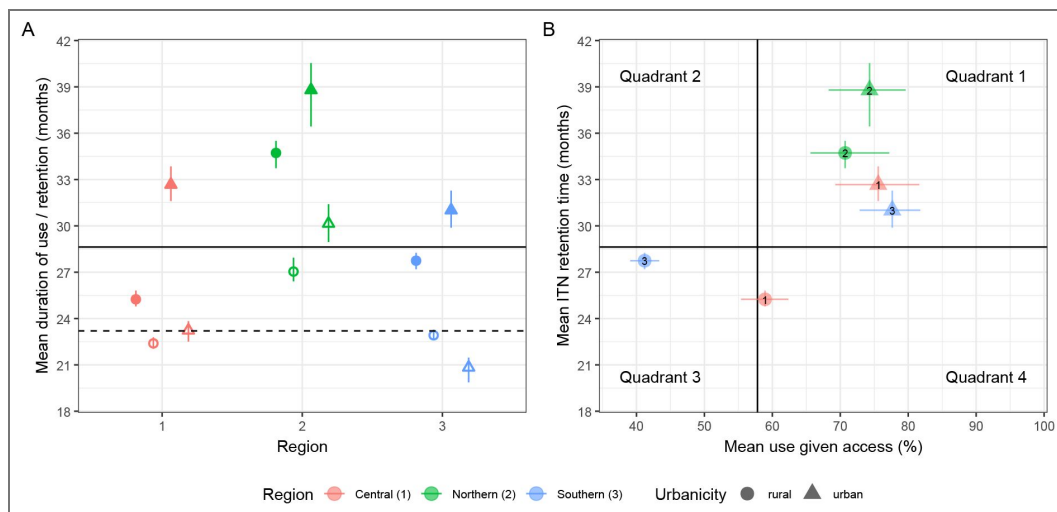


Figure 5—figure supplement 3. Retention time and use given access for Malawi.

All features remain unchanged from figure 5A. Clinical case projections were not conducted for Malawi due to the lack of intervention data stratified to the administrative-one level, despite DHS surveys being conducted at that scale.

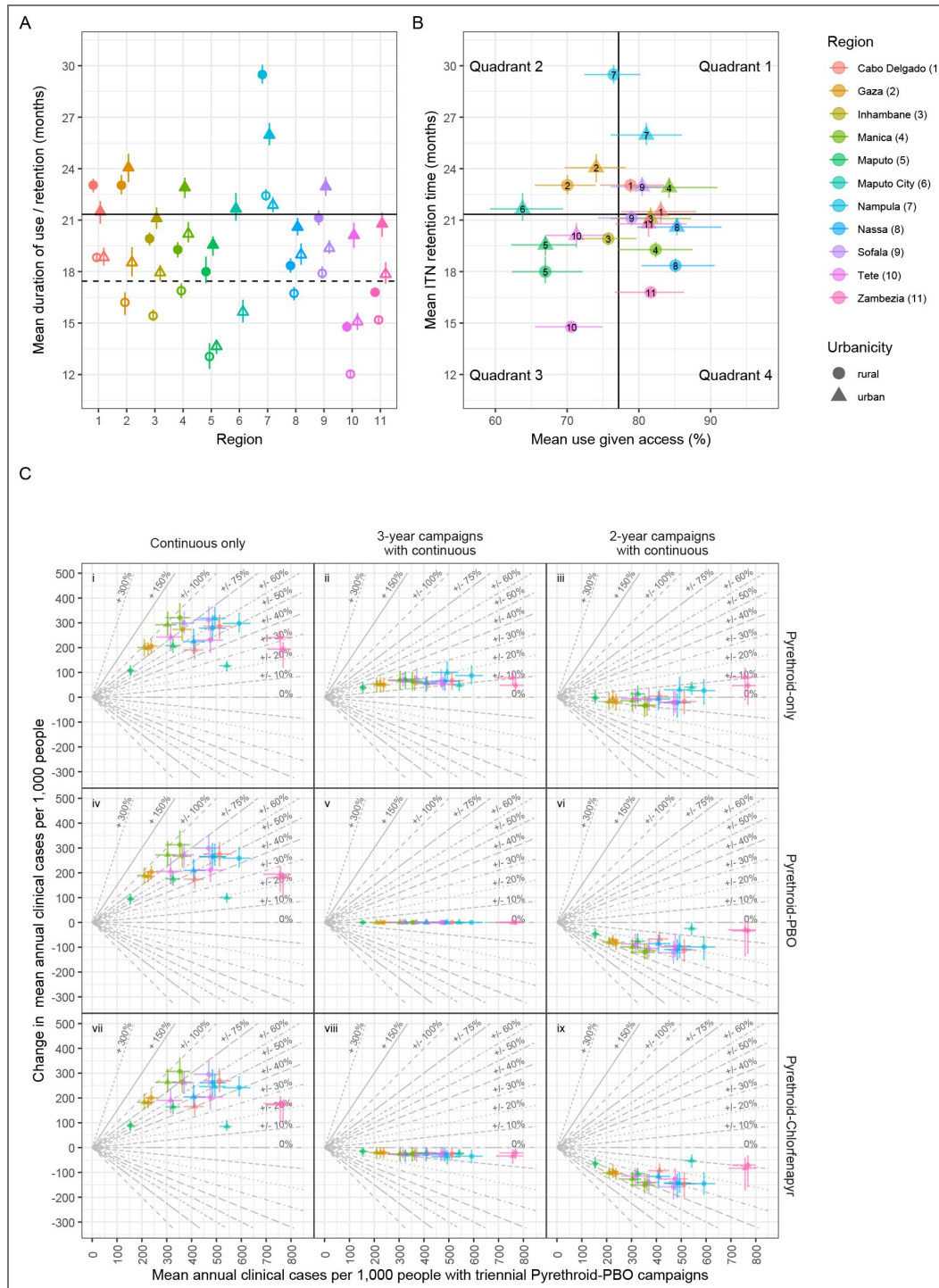


Figure 5—figure supplement 4. Retention time, use given access and changes in cases for Mozambique.

All features remain unchanged from figure 5.

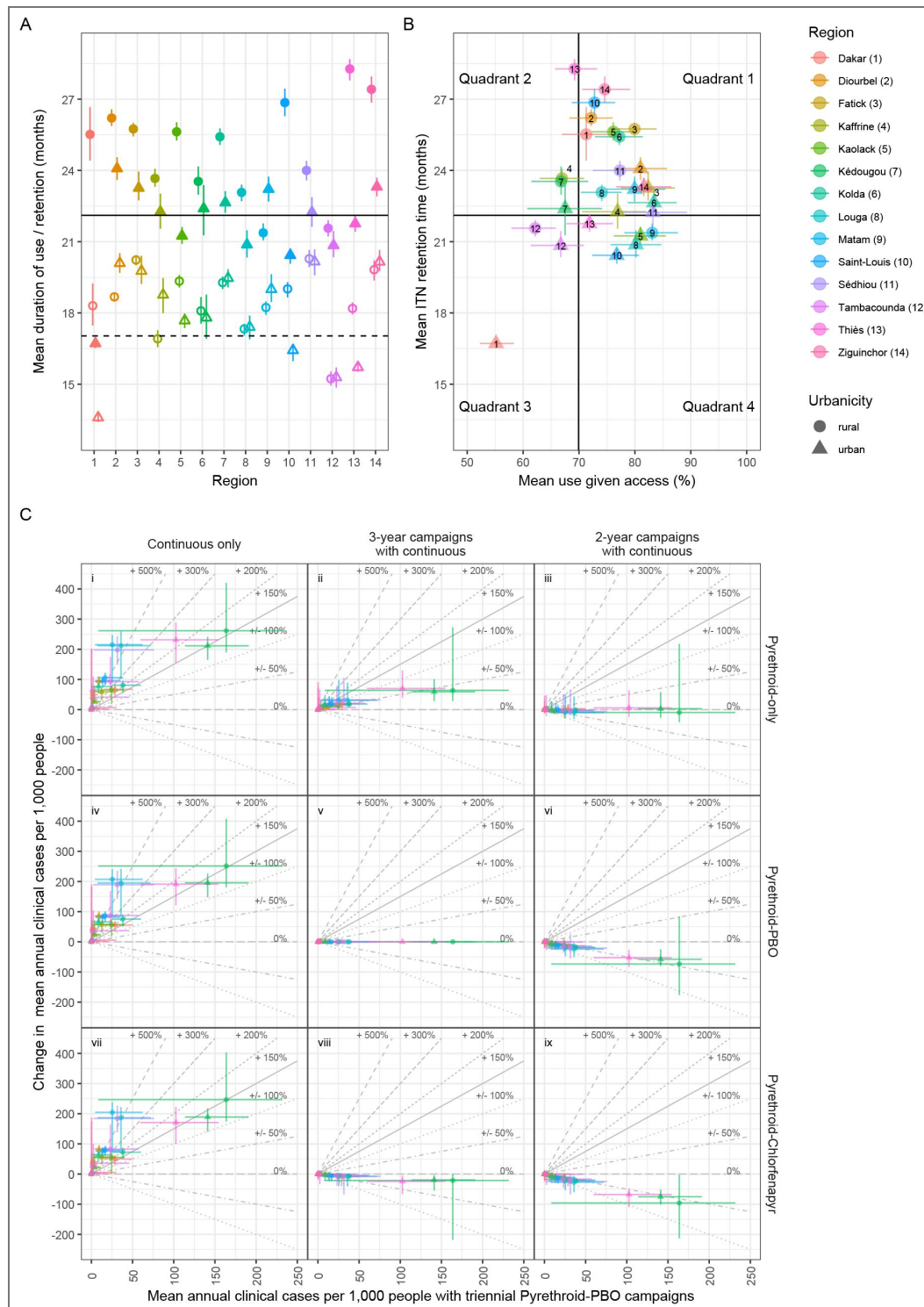


Figure 5—figure supplement 5. Retention time, use given access and changes in cases for Senegal.

All features remain unchanged from figure 5. While use given access was estimated to be broadly similar across urban and rural settings, ITNs were generally estimated to be retained for longer in rural areas;

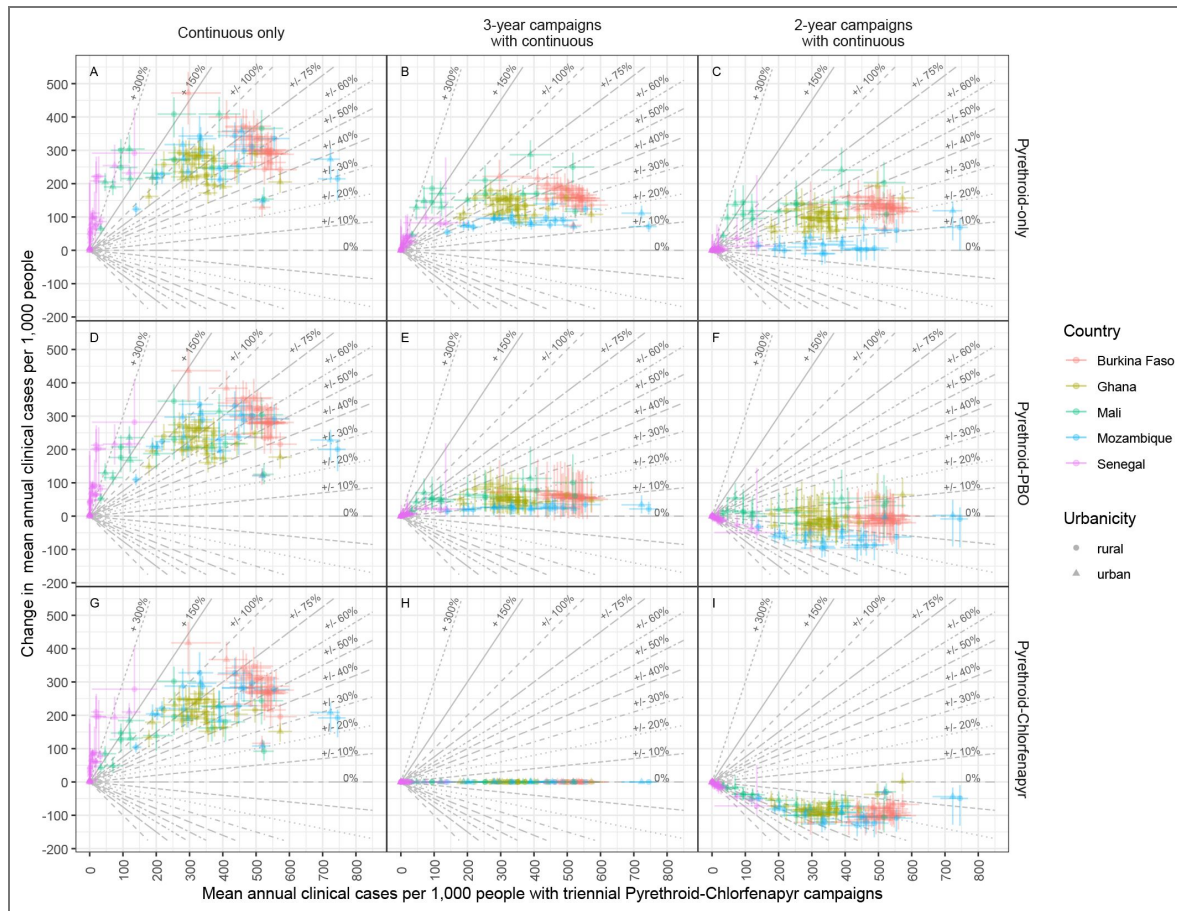


Figure 6—figure supplement 1. Change in cases vs cases with triennial pyrethroidchlorfenapyr distribution.

Points show median estimates of the change in clinical cases following a switch from triennial pyrethroid-chlorfenapyr distribution to alternative strategies against projected clinical cases under the triennial pyrethroid-chlorfenapyr strategy. Figure features otherwise remain unchanged from figure 6 [\[1\]](#).

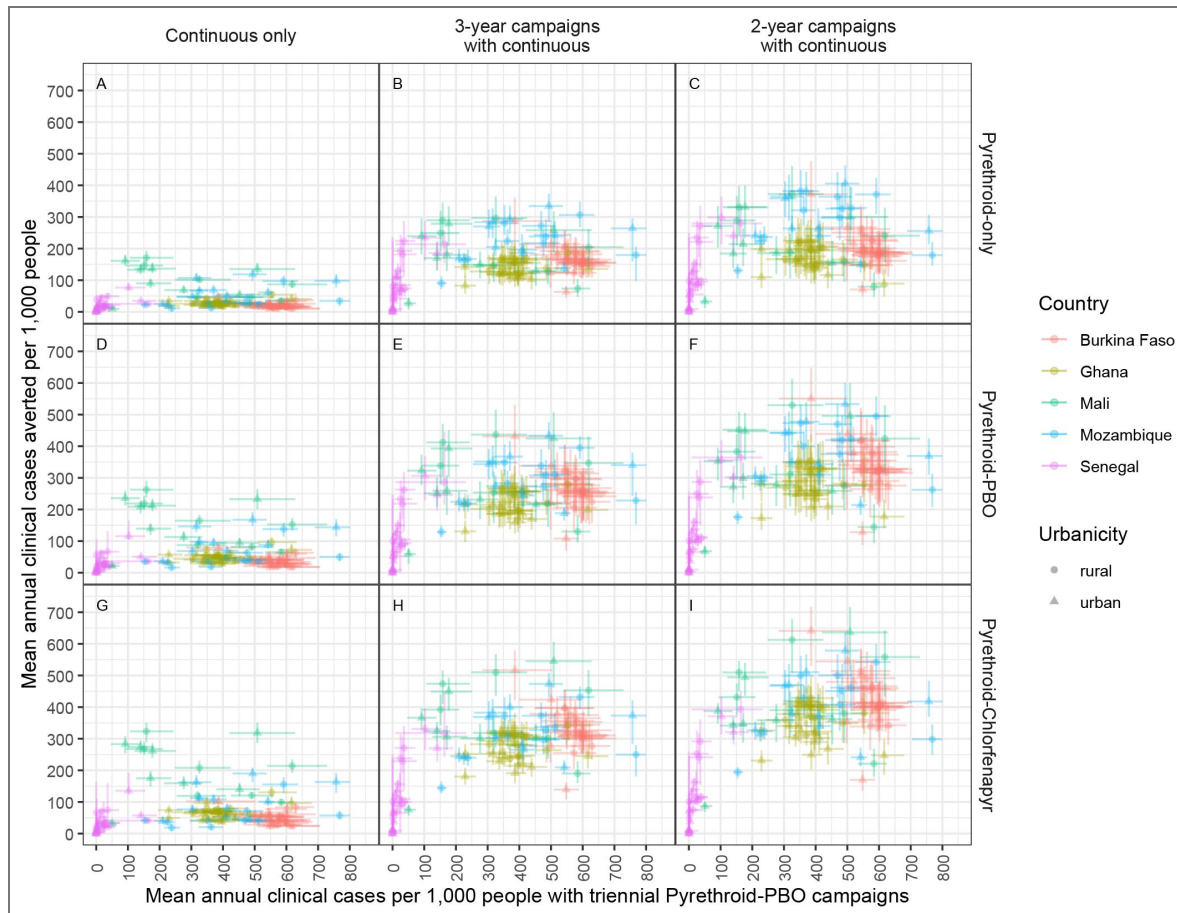


Figure 6—figure supplement 2. Projected cases averted under different ITN distribution strategies.

Points indicate median estimates of the mean annual clinical cases averted by different ITN distribution strategies, in comparison to ceasing ITN distribution entirely, against clinical cases under a triennial pyrethroid-PBO distribution strategy. Figure features otherwise remain unchanged from figure 6 [\[4\]](#).

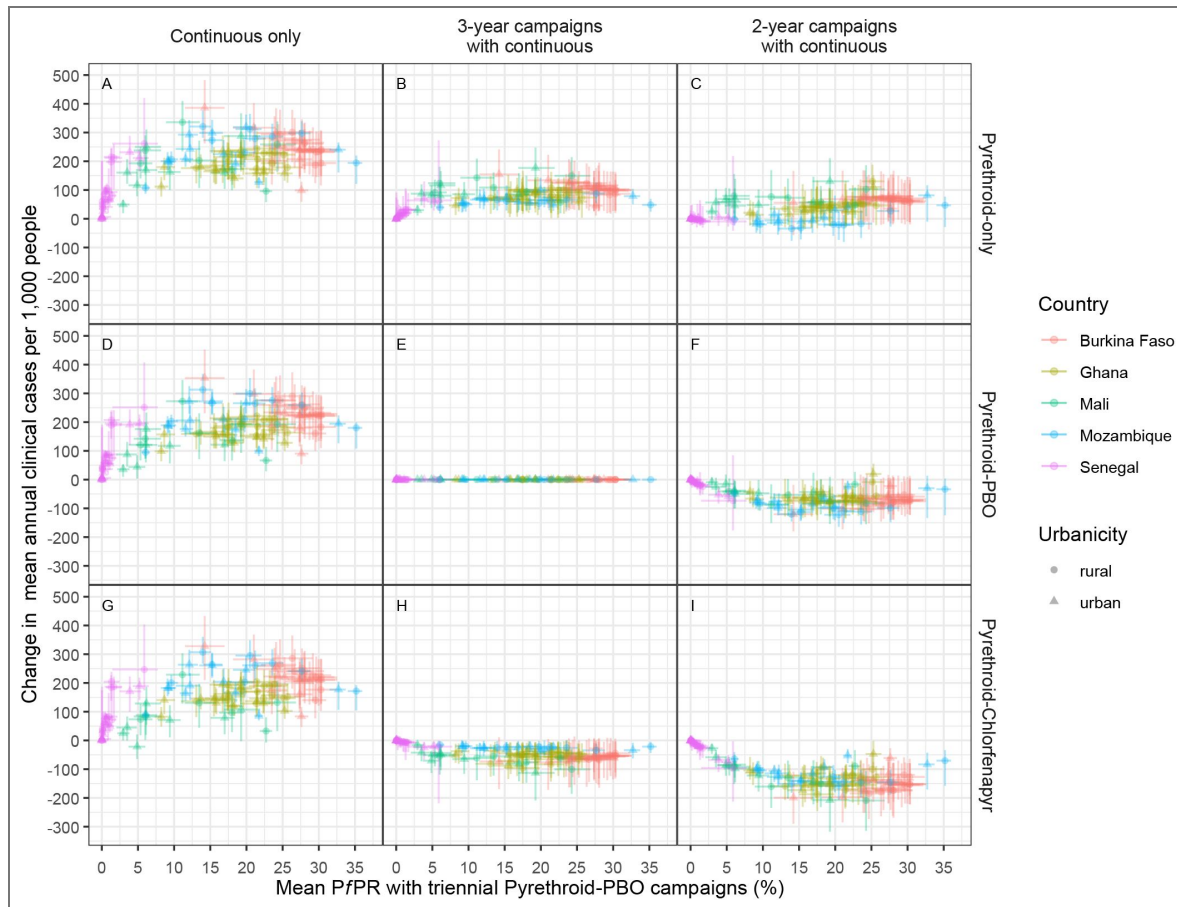


Figure 6—figure supplement 3. Change in cases vs *Pf* PR with triennial pyrethroid-PBO distribution.

Points show median estimates of the change in clinical cases following a switch from triennial pyrethroid-PBO distribution to alternative strategies against all-age *Pf* PR under the triennial pyrethroid-PBO strategy. Figure features otherwise remain unchanged from figure 6 [\[1\]](#).

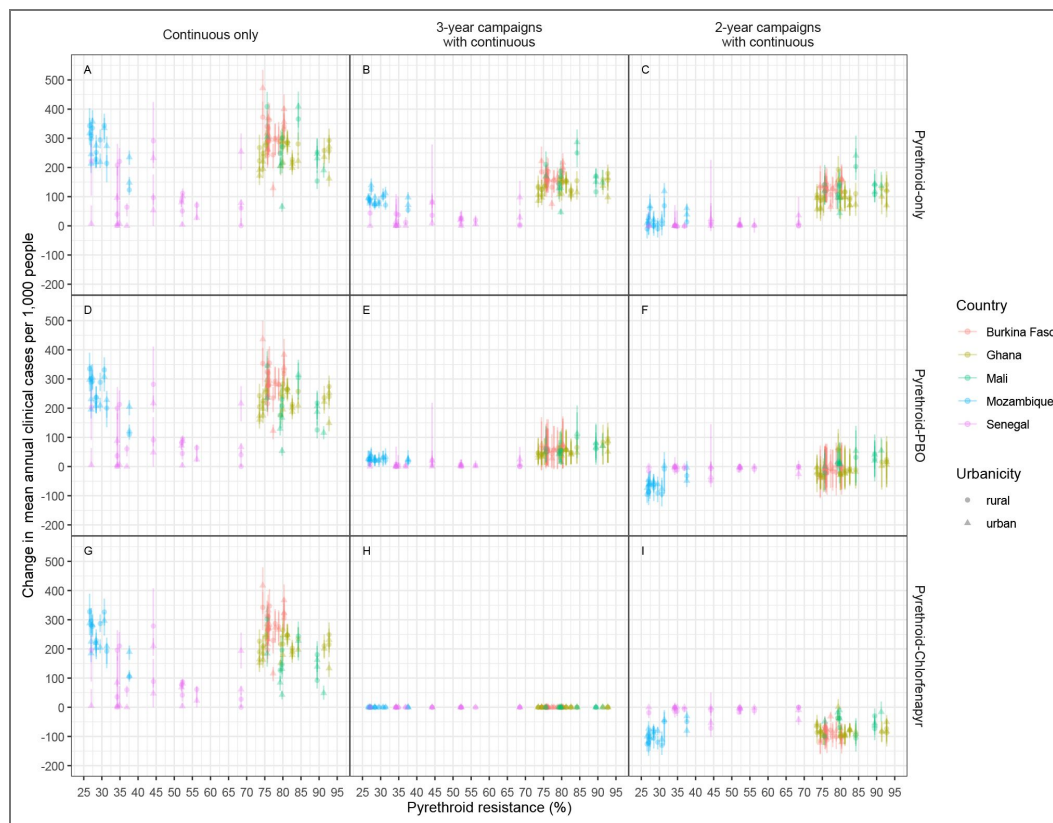


Figure 6—figure supplement 4. Change in cases vs pyrethroid resistance.

Points show median estimates of the change in clinical cases following a switch from triennial pyrethroid-PBO distribution to alternative strategies against pyrethroid resistance. Figure features otherwise remain unchanged from figure 6 [\[link\]](#).

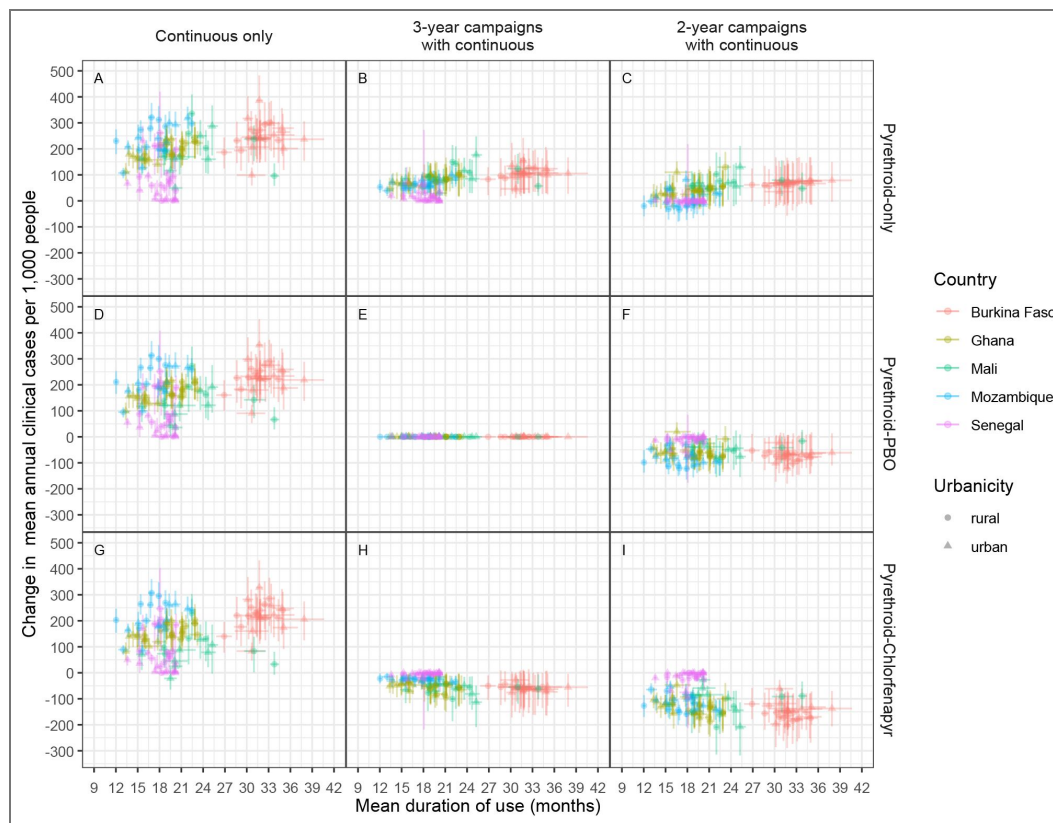


Figure 6—figure supplement 5. Change in cases vs duration of use.

Points show median estimates of the change in clinical cases following a switch from triennial pyrethroid-PBO distribution to alternative strategies against mean duration of use. Figure features otherwise remain unchanged from figure 6 [6](#).

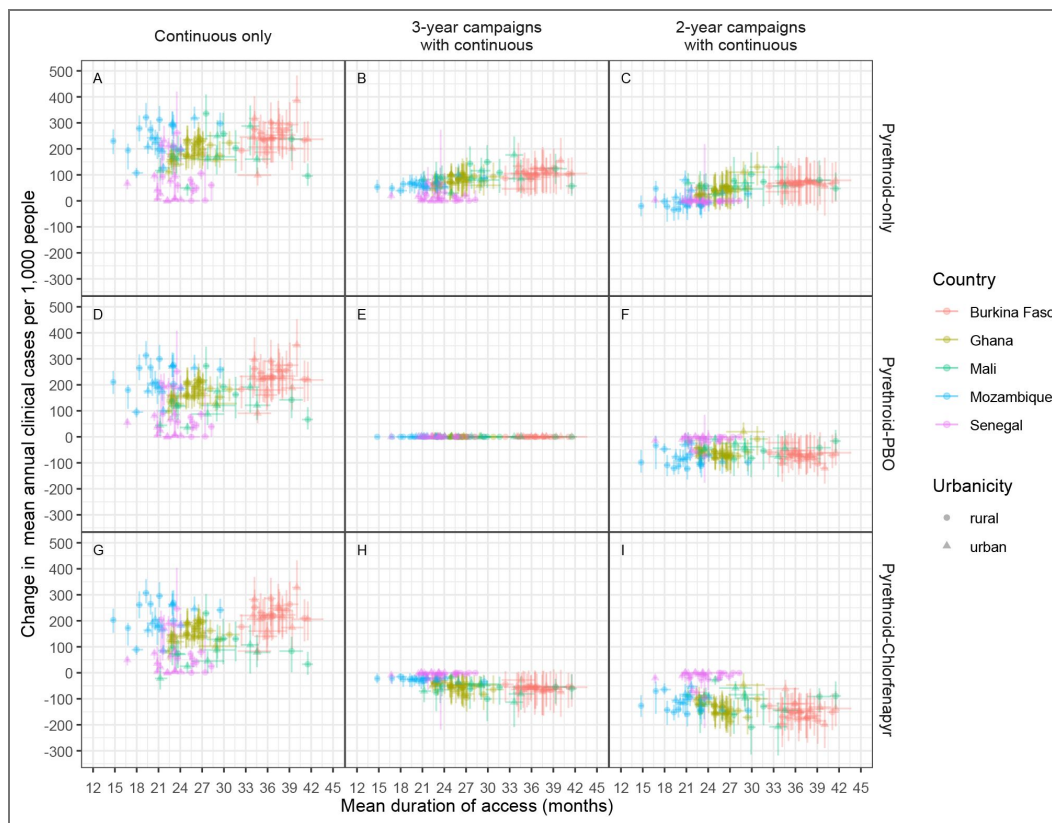


Figure 6—figure supplement 6. Change in cases vs retention time.

Points show median estimates of the change in clinical cases following a switch from triennial pyrethroid-PBO distribution to alternative strategies against mean retention time. Figure features otherwise remain unchanged from figure 6 [\[1\]](#).

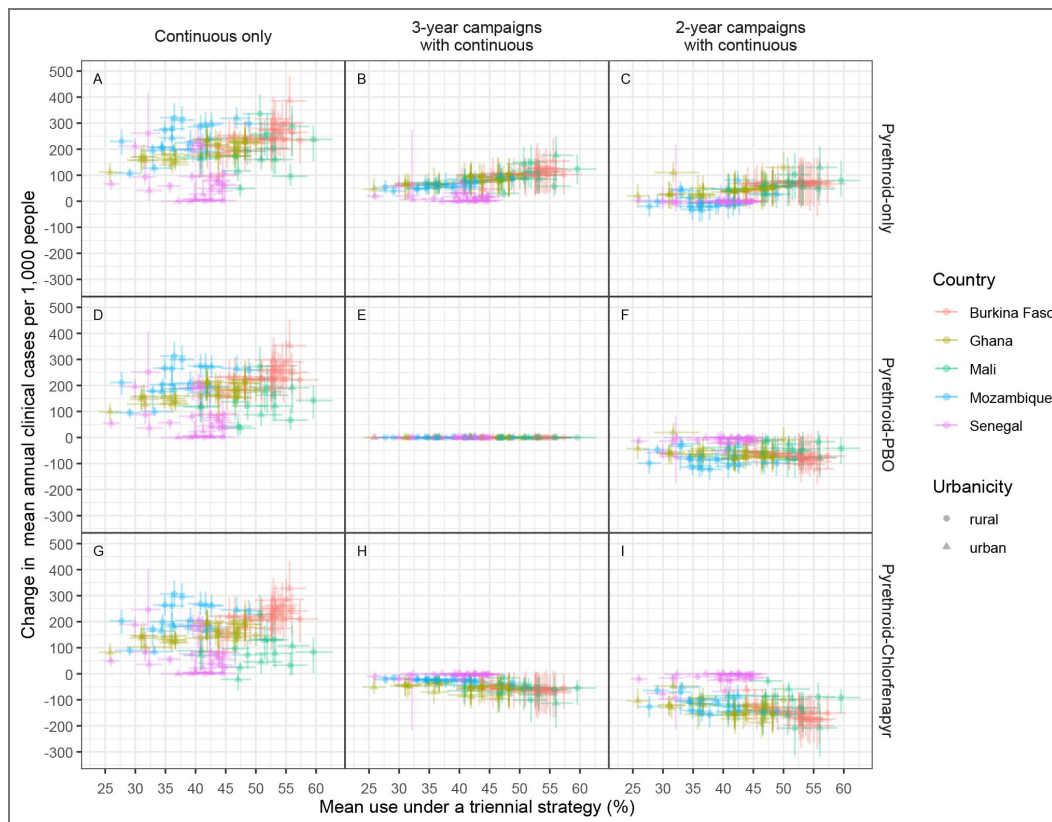


Figure 6—figure supplement 7. Change in cases vs use.

Points show median estimates of the change in clinical cases following a switch from triennial pyrethroid-PBO distribution to alternative strategies against mean use over any triennial distribution strategy. Figure features otherwise remain unchanged from figure 6 [\[1\]](#).

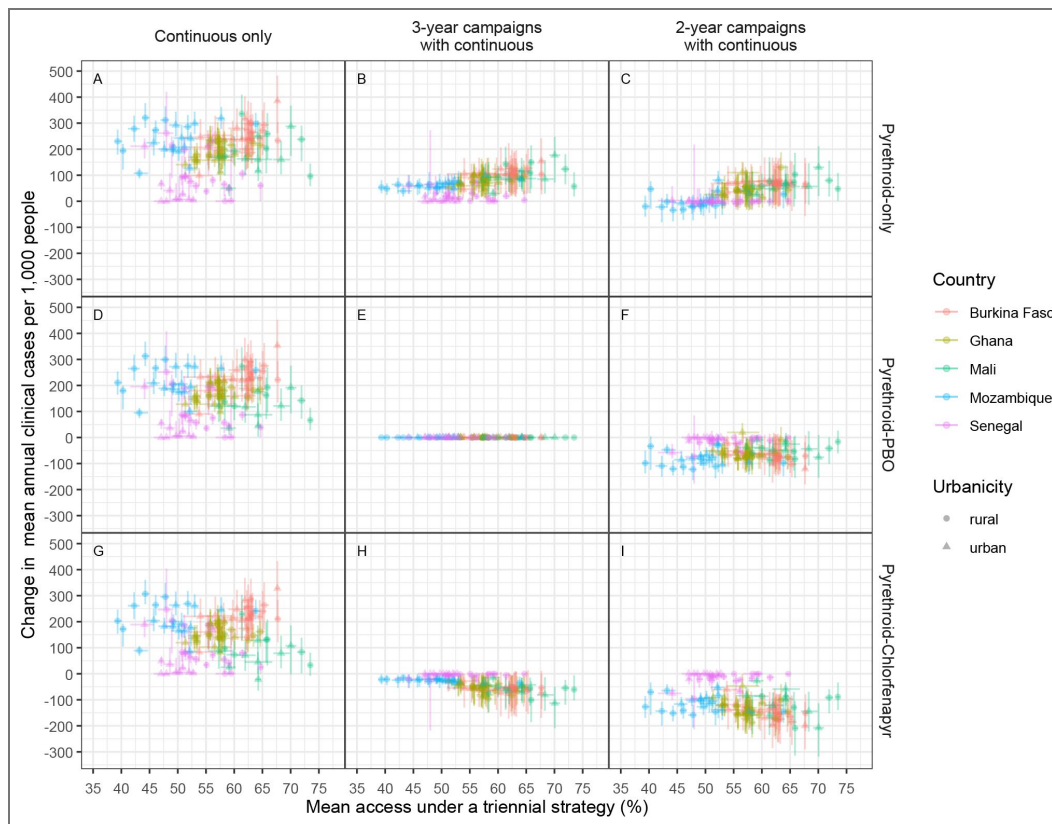


Figure 6—figure supplement 8. Change in cases vs access.

Points show median estimates of the change in clinical cases following a switch from triennial pyrethroid-PBO distribution to alternative strategies against mean access over any triennial distribution strategy. Figure features otherwise remain unchanged from figure 6 [\[1\]](#).

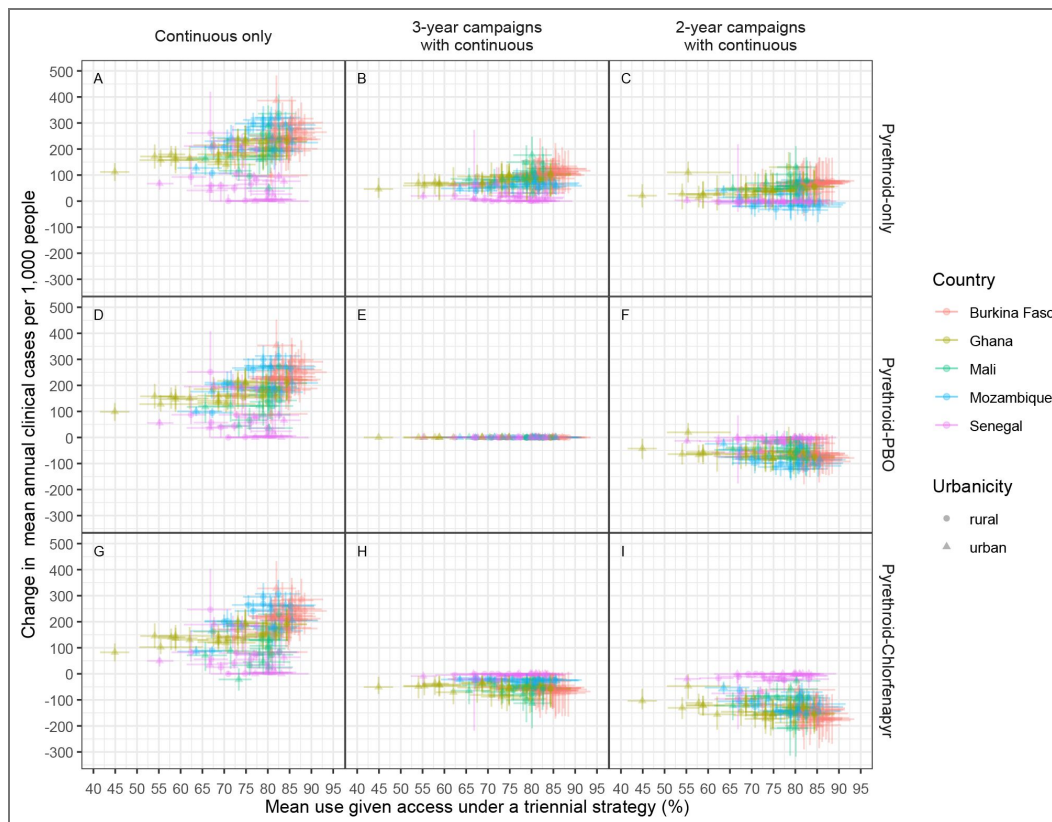


Figure 6—figure supplement 9. Change in cases vs use given access.

Points show median estimates of the change in clinical cases following a switch from triennial pyrethroid-PBO distribution to alternative strategies against mean use given access over any triennial distribution strategy. Figure features otherwise remain unchanged from figure 6 [6](#).

Data Availability

All data used in this study are publicly available. ITN access and use data were obtained from Demographic and Health Surveys (DHS), Malaria Indicator Surveys (MIS), and AIDS Indicator Surveys (AIS), accessible via <https://dhsprogram.com> upon registration. National-level ITN distribution data were sourced from the Alliance for Malaria Prevention (AMP) Net Mapping Project: <https://allianceformalariaprevention.com>. All other covariate data were accessed using the 'site' R package (Winskill et al., 2024), which is publicly available at <https://mrc-ide.github.io/site/>. Country-specific datasets used by the package are accessible upon request to the package maintainers, with instructions provided at the same site. All analyses were conducted in R (v4.3.2). The code for all models and analyses used in this study is publicly available at: https://github.com/andrewcglover/itn_campaigns.

Acknowledgements

The work was funded by the Global Fund under the Net Transition Initiative. ACG and TSC acknowledge funding from the Medical Research Council (MRC) Centre for Global Infectious Disease Analysis (reference number MR/R015600/1), jointly funded by the MRC and the UK Foreign, Commonwealth and Development Office (FCDO), under the MRC/FCDO Concordat agreement, and is also part of the EDCTP2 program supported by the European Union. Authors would like to thank the surveyed communities and those involved in the collection of the Demographic Health System data.

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Author notes

Competing interests: No competing interests declared

Funding Statement: The work was funded by the Global Fund under the Net Transition Initiative. ACG and TSC acknowledge funding from the Medical Research Council (MRC) Centre for Global Infectious Disease Analysis (reference number MR/R015600/1), jointly funded by the MRC and the UK Foreign, Commonwealth and Development Office (FCDO), under the MRC/FCDO Concordat agreement, and as part of the EDCTP2 program supported by the European Union. No additional payments or services were received from a third party for any aspect of the submitted work.

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Peer reviews

Reviewer #1 (Public review):

Summary:

This paper provides a novel method to improve the accuracy of predictions of the impact of ITN strategies, by using sub-national estimates of the duration of ITN access and use over time from cross-sectional survey data and annual country ITNs received.

Strengths:

The approach is novel, makes use of available data, and has considered all of the relevant components of ITN distributions.

Weaknesses:

(1) The main message of the paper was not very clear, and did not seem to fit the title. The title focuses on sub-national tailoring of ITN, but the abstract did not feature results directly about SNT. It was not very clear what the main result of the paper was - there are several ITN observations in the results and discussion. Most did not seem to be directly about SNT, but rather sub-national differences in use and access were accounted for in the analyses. It was not clear if the same conclusions would be reached without accounting for sub-national differences, but the estimates and predictions could be expected to be more accurate.

(2) Some of the results seemed to me to be apparent even without a modelling exercise (eg high coverage could not be maintained between campaigns, use would be higher with 2-yearly distributions rather than 3-yearly) or were not in themselves new insights (eg estimates of the duration of use). It would be helpful to clearly state what the novel results are in the abstract, the first paragraph of the discussion and the conclusions, and to make sure that the title is consistent.

(3) On L236, the link to SNT is stated: "the models indicate trends that can support sub-national tailoring of ITNs". They could indeed, but SNT itself is not done in this paper. It seems to be about improving sub-national predictions of the impact of single ITN strategies, by taking into account sub-national variation in access and use duration. This is useful, and the model developed has novel aspects.

(4) Individual countries may have records on when nets were distributed to the regions rather than needing to use the annual country number of nets together with the DHS data. It could be helpful to say what the analysis steps would be in that case.

(5) There were several assumptions that needed to be made in building the model. There is some validation of the timing of the distributions (L633 "verified where possible through discussion with interested parties nationally and internationally") and the fit of estimated access and use to survey data, and agreement between predictions of prevalence and MAP estimates. It would be helpful to say which assumptions are important for the results (and would be key knowledge gaps) and which would not make a difference. It might be possible to validate the net timing model using a country where net distributions are known reasonably well.

(6) What was assumed about what happens to old nets after a mass campaign was not clear. This assumption is likely to affect the predictions of access for the biennial distributions.

(7) L312 and elsewhere: That use given access declines with net age is plausible. However, I wondered if this could be partly a consequence of the assumptions in the model (eg the two exponential decays for access and use, the possible assumption that new nets displace the current ones when there is a mass campaign).

(8) The Methods section on Estimating historical use and access seemed to be aimed at readers familiar with formulae, but I think it could lose other interested readers. It could be useful to explain a little more about what is happening at each step and also why.

(9) The model was fitted to MAP estimates of PfPR2-10, which themselves come from a model. It may be that there is different uncertainty in the MAP estimates for different regions. I couldn't see this on the graph, but maybe the uncertainty is small. Was this taken into account in the fitting?

(10) Was uncertainty from each estimated component integrated into the other components?

(11) Eyeballing Figure 2 (Burkina Faso), there is a general pattern of decline in all the regions, some differences between the regions and some differences in how well the model fits between the regions. If possible, it could be helpful to say how much better the fit was when using region-specific compared to countrywide parameter values for access and use, and how different the results would be.

(12) The question of moving from a campaign every three to every two years may not be the most pertinent question in the current funding landscape. I realise that a paper is in development for a long time, but it would be helpful to comment on what else the model could be used for when fewer rather than more nets are likely to be available.

<https://doi.org/10.7554/eLife.108745.1.sa2>

Reviewer #2 (Public review):

Summary:

The authors design a custom Bayesian model to estimate the probabilities of access, use and use given access of insecticide-treated nets in six African countries, providing sub-national estimates and inferring the average duration of ITN use and access. An individual-based model was employed to simulate malaria epidemics and estimate the effectiveness of different ITN distribution strategies. The study finds that the mean probability of use or access did not reach 80% (a universal coverage formerly targeted by WHO) for any of the regions, even for biennial campaigns, demonstrates that switching from triennial to biennial distribution campaigns increases population use by 7.9%, and evaluates the impact of employing more efficient ITNs on *P. falciparum* prevalence.

Strengths:

The authors developed a data-driven model that accounts for data collection imperfections and sources of uncertainty while differentiating between ITN use and access. They developed a methodology to infer the timing of a mass campaign from publicly available data instead of assuming fixed dates. The probability of use given access allows for determining the regions where ITN distribution is least effective. This work can help better inform future interventions by identifying regions where increasing mass campaign frequency or employing better ITNs are most effective. Finally, in addition to insights on ITN access and use for the six countries analyzed, the paper contributes a methodological framework that can likely be extended to other countries.

Weaknesses:

Since the models employed are rather complex, the description of the methodology may be hard to follow for most readers. In addition, the models assume many hypotheses, including:

- (1) Exponential decay of ITN use/access.
- (2) The decay rates for the probability of the ITN repelling and killing a mosquito are the same.
- (3) Given a time instant, all individuals in the same administrative unit and have the same probability of using a net;
- (4) ITN use/access decay models do not depend on the distribution strategy (e.g. biennial vs triennial distribution).
- (5) The Bayesian model assumes some narrow prior distributions.

The impact of these hypotheses on the estimated parameters is not explored in the paper, and no sensitivity analyses are performed, although some limitations are discussed.

<https://doi.org/10.7554/eLife.108745.1.sa1>

Author response:

We would like to thank both reviewers for taking the time to review the manuscript in detail. Your comments have been extremely useful and constructive. A revised version of the manuscript will seek to address the weaknesses raised, clarifying the reasons for the assumptions made, the impact they have and how they influence the policy implication of the work. We will clarify the language to differentiate the work from the standard sub-national tailoring which is typically conducted to support National Malaria Programmes and emphasise why our mechanistic model can provide greater information than simple summary statistics.

<https://doi.org/10.7554/eLife.108745.1.sa0>